

A Memoir on the Transformation of Elliptic Functions

(Paper 578, continued from p. 142 — pp. 143–175 of Vol. IX)

[p. 143] § 33 (continued). The modular equation in its six forms

Thus $n = 3$, starting from $\frac{1}{M} = 1 + \frac{2v^2}{v}$, we have

$$\frac{1}{M} = 1 + \frac{2v^2}{v}, \quad -3M = 1 - \frac{2v^2}{u}, \quad \frac{v^2}{u^2M} = 1 + \frac{2v}{u^2}, \quad -\frac{u^2}{v^2} 3M = 1 - \frac{2u}{v^2};$$

and of course if from any two of these we eliminate M , we have either an identity or the modular equation; thus we have the modular equation under the six different forms:

$$\begin{aligned} (1, 2) \quad & (v + 2u^2)(u - 2v^2) + 3uv = 0, \\ (1, 3) \quad & v^2(v + 2u^2) - u(u^2 + 2v) = 0, \\ (1, 4) \quad & (v + 2u^2)(v^2 - 2u) + 3u^2 = 0, \\ (2, 3) \quad & (u - 2v^2)(u^2 + 2v) + 3v^2 = 0, \\ (2, 4) \quad & v(v^2 - 2u) - u^2(u - 2v^2) = 0, \\ (3, 4) \quad & (u^2 + 2v)(v^2 - 2u) + 3uv = 0. \end{aligned}$$

34. Next $n = 5$. Here, starting from $\frac{1}{M} = \frac{v - u^2}{v(1 - uv^2)}$, the changes give

$$\frac{1}{M} = \frac{v - u^2}{v(1 - uv^2)}, \quad 5M = \frac{u + v^2}{u(1 + u^2v)}, \quad \frac{v^2}{u^2M} = \frac{v^2(v - u^2)}{u^2(1 - uv^2)}, \quad \frac{u^2}{v^2} 5M = \frac{u^2(u + v^2)}{v^2(1 + u^2v)};$$

viz. the third and the fourth forms agree with the first and the second forms respectively; that is, there are only two independent forms, and the elimination of M from these gives

$$5uv(1 - uv^2)(1 + u^2v) - (v - u^2)(u + v^2) = 0,$$

which is a form of the modular equation.

35. In the case $n = 7$, starting from $\frac{1}{M} = \frac{-7u(1 - uv)(1 - uv + u^2v^2)}{u - v^2}$ (as to this see post, No. 43), the forms are

$$\frac{1}{M} = \frac{-7u(1 - uv)(1 - uv + u^2v^2)}{u - v^2}, \tag{1}$$

$$-7M = \frac{-7v(1 - uv)(1 - uv + u^2v^2)}{v - u^2}, \tag{2}$$

$$\frac{v^2}{u^2M} = \frac{-7v^2(1 - uv)(1 - uv + u^2v^2)}{u^2(v - u^2)}, \tag{3}$$

$$-\frac{u^2}{v^2} 7M = \frac{-7u^2(1 - uv)(1 - uv + u^2v^2)}{v^2(u - v^2)}; \tag{4}$$

so that here again the third and the fourth forms are identical with the second and the third forms respectively; there are thus only two forms, and the elimination of M gives

$$(u - v^2)(v - u^2) + 7uv(1 - uv)(1 - uv + u^2v^2) = 0,$$

which is a form of the modular equation.

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36. If in the foregoing equation

$$F'v \cdot \frac{1}{M} = R(u, v),$$

we make the change $u, v, \frac{1}{M}$ into $v, \pm u, \pm nM$, it becomes

$$\pm nM \cdot F'u = R(v, \pm u);$$

combining these equations, we have

$$\pm nM^2 \cdot \frac{F'u}{F'v} = \frac{R(v, \pm u)}{R(u, v)};$$

or, substituting herein the foregoing value

$$M^2 = -\frac{1}{n} \frac{(1-v^2)vF'v}{(1-u^2)uF'u},$$

this becomes

$$\mp \frac{v(1-v^2)}{u(1-u^2)} = \frac{R(v, \pm u)}{R(u, v)} \quad \begin{array}{l} + \text{ for } n \equiv 3 \text{ or } 5 \pmod{8}, \\ - \text{ for } n \equiv 1 \text{ or } 7 \pmod{8}, \end{array}$$

which must agree with the modular equation. Thus in the last-mentioned case $n = 3$, we have

$$\frac{1}{2}F'v \cdot \frac{1}{M} = 3(v^2u^2 + 2u^2v + 1)u,$$

or, say

$$R(u, v) = +3(v^2u^2 - 2u^2v + 1)u, \quad \text{and therefore} \quad R(v, -u) = (v^2u^2 - 2uv^2 + 1)v;$$

the equation is

$$\frac{v(1-v^2)}{u(1-u^2)} = \frac{(v^2u^2 - 2uv^2 + 1)v}{(v^2u^2 - 2u^2v + 1)u},$$

which is right; because Jacobi, p. 82, [*Ges. Werke*, t. I., p. 137], for the modular equation, gives

$$1 - u^4 = (1 - u^2v)(v^2u^2 + 2u^2v + 1), \quad 1 - v^4 = (1 - uv^2)(v^2u^2 - 2uv^2 + 1).$$

Observe that the general equation

$$\mp \frac{v(1-v^2)}{u(1-u^2)} = \frac{R(v, \pm u)}{R(u, v)}$$

no longer contains the functions $F'v$, $F'u$, which enter into Jacobi's expression of M^2 .

Theorem in connexion with the multiplication of Elliptic Functions. Art. Nos. 37 to 40.

37. The theory of multiplication gives an important theorem in regard to transformation. Starting with the n th transformation

$$\frac{1-y}{1+y} = \frac{1-x}{1+x} \left(\frac{\alpha - \beta x + \gamma x^2 - \dots}{\alpha + \beta x + \gamma x^2 + \dots} \right)^2, \quad = \frac{1-x}{1+x} \left(\frac{P-Qx}{P+Qx} \right)^2,$$

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we may form a like transformation,

$$\frac{1-z}{1+z} = \frac{1-y}{1+y} \left(\frac{\alpha' - \beta'y + \gamma'y^2 - \dots}{\alpha' + \beta'y + \gamma'y^2 + \dots} \right)^2, \quad = \frac{1-y}{1+y} \left(\frac{P'-Q'y}{P'+Q'y} \right)^2,$$

such that the combination of the two gives a multiplication, viz. for the relation between y , z , deriving w from v as v from u , we have $w = u$; and instead of M we have $M' = \pm \frac{1}{nM}$; that is, we have

$$\frac{dx}{\sqrt{1-x^2 \cdot 1-u^2x^2}} = \frac{M dy}{\sqrt{1-y^2 \cdot 1-v^2y^2}},$$

$$\frac{dy}{\sqrt{1-y^2 \cdot 1-v^2y^2}} = \frac{M' dz}{\sqrt{1-z^2 \cdot 1-u^2z^2}},$$

and thence

$$\frac{dx}{\sqrt{1-x^2 \cdot 1-u^2x^2}} = \frac{\pm \frac{1}{n} dz}{\sqrt{1-z^2 \cdot 1-u^2z^2}},$$

or, writing $x = \text{sn } \theta$, we have $z = \pm \text{sn } n\theta$; $\frac{n-1}{n}$ is here $(-)^{\frac{n-1}{2}}$, viz. it is $-$ for $n \equiv 3$ or $7 \pmod{8}$, and $+$ for $n \equiv 1$ or $5 \pmod{8}$.

Now in part effecting the substitution, we have

$$\frac{1-z}{1+z} = \frac{1-x}{1+x} \left(\frac{P-Qx}{P+Qx} \right)^2 \left(\frac{P'-Q'y}{P'+Q'y} \right)^2,$$

where y denotes its value in terms of x .

And from the theory of elliptic functions, replacing $\operatorname{sn} n\theta$, $\operatorname{sn} \theta$ by their values $\pm z$, x , we have an equation

$$\frac{1-z}{1+z} = \frac{1-x}{1+x} \left(\frac{A-Bx+Cx^2-\dots}{A+Bx+Cx^2+\dots} \right)^2,$$

where $A - Bx + Cx^2 - \dots$, $A + Bx + Cx^2 + \dots$ are given functions each of the order $\frac{1}{2}(n^2 - 1)$; viz. the coefficients are given functions of k , or, what is the same thing, of u^4 .

Comparing the two results, we see that in the n th transformation the sought-for function, $\alpha + \beta x + \gamma x^2 + \dots$ of the order $\frac{1}{2}(n - 1)$, is a factor of a given function $A + Bx + Cx^2 + \dots$ of the order $\frac{1}{2}(n^2 - 1)$.

38. Considering the modular equation as known, then by what precedes we have

$$\frac{1}{\alpha + \beta x + \gamma x^2 + \dots} = \frac{1}{a} \left(1 + \frac{B}{a} x + \dots + \frac{w^n}{a} x^{\frac{n-1}{2}} \right);$$

that is, the given function $A + Bx + Cx^2 + \dots$ has a factor $1 + \frac{B}{a} x + \dots + \frac{w^n}{a} x^{\frac{n-1}{2}}$, of which one (the last) coefficient is known, and we are hence able theoretically to

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determine all the other coefficients rationally in terms of u , v ; that is, the modular equation being known, we can theoretically complete the solution of the transformation problem. I do not, however, see the way to obtaining a convenient solution in this manner.

39. The formula in question for $n = 3$ is

$$\frac{1 + \operatorname{sn} 3\theta}{1 - \operatorname{sn} 3\theta} = \frac{1 + \operatorname{sn} \theta}{1 - \operatorname{sn} \theta} \left(\frac{1 + 2 \operatorname{sn} \theta - 2k^2 \operatorname{sn}^3 \theta - k^2 \operatorname{sn}^4 \theta}{1 - 2 \operatorname{sn} \theta + 2k^2 \operatorname{sn}^3 \theta - k^2 \operatorname{sn}^4 \theta} \right)^2,$$

which, putting therein $x = \operatorname{sn} \theta$, $z = -\operatorname{sn} 3\theta$, and replacing k by u^4 , may be written

$$\frac{1-z}{1+z} (\div) = (1+x)(1-2x+2u^2x^3-u^4x^4)^2 (\div),$$

where the signs $(\div)$ indicate denominators which are obtained from the numerators by changing the signs of z , x respectively.

The theorem in regard to $n = 3$ thus is, $1 + \frac{v^2}{u^2} x$ is a factor of $1 - 2x + 2u^2x^3 - u^4x^4$; viz. writing in the last-mentioned function $x = -\frac{u^2}{v^2}$, we ought to have

$$0 = 1 + \frac{2u^2}{v^2} + \frac{2u^8}{v^6} - \frac{u^4}{v^4} \cdot \frac{u^8}{v^8},$$

that is,

$$u^4 + 2uv - 2u^2v^3 - v^6 = 0,$$

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[Table showing the polynomial multiplication $f \cdot g$ “Term in [] has factor $1 + \frac{\beta}{\alpha}x + \frac{w^2}{v}x^2$ ” on the left and the expanded sum of $f \cdot g$ on the right, organised by powers of x . Cayley simply writes out the formal product; we render the two columns as paired displays.]

Left column (factors and partial products):

$$\text{Term in [] has factor } 1 + \frac{\beta}{\alpha}x + \frac{w^2}{v}x^2;$$

$$n = 1, \text{ term in [] is } = (1+x)^2(1-x)^6.$$

Right column (sum, by ascending powers of x):

$$\begin{aligned}
 &+ 56u^{20}(1 + 2u^4) x^{15} \\
 &+ 8u^{16}(46 + 57u^4 + 8w^{20}) x^{14} \\
 &- 8u^{14}(16 + 25u^4 + 4w^{20}) x^{13} \\
 &- u^{12}(16 + 305u^4 + 144w^{20}) x^{12} \\
 &+ 4u^{10}(8 + 51u^4 + 16w^{20}) x^{11} \\
 &+ 28u^{10}(1 + 4u^4) x^{10} \\
 &- 28u^8(2 + 3u^4) x^9 \\
 &- 14u^6 x^8 \\
 &+ 4u^6(7 + 2u^4) x^7 \\
 &- 4u^4 x^6 \\
 &- 4u^4 x^5 \\
 &+ u^4 x^4 \\
 &+ (\div).
 \end{aligned}$$

Term in [] has factor

$$1 + \frac{\beta}{\alpha}x + \frac{w^2}{v}x^2;$$

$u = 1$, term in [] is $= (1 + x)^2(1 - x)^6$.

The transformations $n = 3, 5, 7, 11$. Art. Nos. 41 to 51.

41. The cubic transformation, $n = 3$.

I reproduce the results already obtained; since there are only two coefficients α, β , these are also the last but one and last coefficients ρ, σ . Hence, from the values of $\alpha, \beta, \rho, \sigma$, we have

$$\alpha = 1, \quad 2\alpha = \frac{v^2}{u} \left(\frac{1}{M} - \frac{u^2}{v^2} \right), \quad 2\beta = \frac{1}{M} - 1, \quad \beta = \frac{u^2}{v};$$

the two values of $\frac{1}{M}$ are thus $\frac{1}{M} = 2\frac{u}{v^2} + \frac{u^2}{v^2}, = 1 + \frac{2u^2}{v}$, giving the modular equation

$$v^4 + 2u^2v^2 - 2uv - u^4 = 0,$$

and we then have

$$\frac{1-y}{1+y} = \frac{1-x}{1+x} \left(\frac{v-u^2x}{v+u^2x} \right)^2.$$

[p. 148] **42. The quintic transformation, $n = 5$.**

Here there are the three coefficients α, β, γ , or β, γ are the last but one and last coefficients ρ, σ ; we have

$$\alpha = 1, \quad 2\beta = v^2u \left(\frac{1}{M} - \frac{u^4}{v^2} \right), \quad 2\beta = \frac{1}{M} - 1, \quad \gamma = \frac{u^2}{v}.$$

Comparing the two values of β , we have $\frac{1}{M} = \frac{v - u^2}{v(1 - uv^2)}$, and then

$$\alpha = 1, \quad 2\beta = \frac{u(v^2 - u^2)}{v(1 - uv^2)}, \quad \gamma = \frac{u^2}{v},$$

so that only the modular equation remains to be determined.

The unused equation is

$$2\alpha\gamma + 2\alpha\beta + \beta^2 = \frac{u^2}{v^2}(2\alpha\gamma + 2\beta\gamma + \beta^2),$$

which, putting therein $\alpha = 1$, may be written

$$u^4(\gamma^2 + 2u\gamma + u^2) = 2\beta(\gamma u^2 - v^2);$$

attending to the value of β , this divides by $u^2 - v^2$; in fact the equation may be written

$$2\gamma + \beta^2 = -\frac{u(v^2+u^2)}{v(1-uv^2)}(\gamma u^2 - v^2);$$

and then completing the substitution, and integralising, this becomes

$$[8uv^2(1-uv^2) + (v^2 - u^2)^2] = 4uv(u^2 + v^2)(1-uv^2)(1-uv^2),$$

viz.

$$4(1-uv^2)uv = 2u^4(1-u^2)(v^2+u^2)(1-uv^2)(1-uv^2) + (v^2-u^2)^2 \cdot u^2(1-u^2v^2)(1-u^2v^2) = 0;$$

and the term in [] being $-(v^2 - u^2)(1-uv^2)(1-uv^2)$, the whole again divides by $v^2 - u^2$, and the equation thus becomes

$$(v^2 + u^2)(v^2 - u^2) - 4uv(1-uv^2)(1-uv^2 + u^2v^2) = 0,$$

which is the modular equation.

43. The septic transformation, $n = 7$.

I do not propose to complete the solution directly from the fundamental equations for $\alpha, \beta, \gamma, \delta$, but resort to the known modular equation, and to an expression of M which I obtain by means thereof.

The modular equation is

$$(1-u^2)(1-v^2) - (1-uv)^4 = 0,$$

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which may also be written

$$(v-u^2)(u-v^2) + 7uv(1-uv)(1-uv+u^2v^2) = 0,$$

as can be at once verified; but it also follows from Cauchy's identity

$$(x+y)^7 - x^7 - y^7 = 7xy(x+y)(x^2+xy+y^2)^2.$$

We then have

$$M = -\frac{1}{n} \frac{(1-v^2)vF'v}{(1-u^2)uF'u}.$$

Moreover

$$\begin{aligned} uF'u &= -u^2(1-v^2) + uv(1-uv)^2 \\ &= -\frac{u^2}{1-u^2}(1-uv)^2 + uv(1-uv)^2 \\ &= \frac{(1-uv)^2}{1-u^2} u(v-u^2); \end{aligned}$$

and similarly

$$F'v = \frac{(1-uv)^2}{1-v^2} v(u-v^2);$$

whence

$$\frac{1}{M} = -\frac{7u}{v} \frac{(v-u^2)}{u-v^2}.$$

Writing this under the form

$$\frac{1}{M^2} = \frac{-7uv(v-u^2)(u-v^2)}{(u-v^2)^2}, \quad = \frac{49u^2(1-uv)^2(1-uv+u^2v^2)^2}{(u-v^2)^2},$$

I find, as will appear, that the root must be taken with the sign $-$, and that we thus have $\frac{1}{M} = \frac{-7u(1-uv)(1-uv+u^2v^2)}{u-v^2}$, whence also $M = \frac{v(1-uv)(1-uv+u^2v^2)}{u-v^2}$.

44. Recurring now to the fundamental equations for the septic transformation, the coefficients are $\alpha, \beta, \gamma, \delta$, and we have

$$\alpha = 1, \quad 2\gamma = v^2u\left(\frac{1}{M} - \frac{u^6}{v^2}\right),$$

$$2\beta = \frac{1}{M} - 1, \quad \delta = \frac{u^2}{v},$$

so that the coefficients are all given in terms of v , M . The unused equations are

$$u^2(3\alpha\gamma + 2\alpha\delta + \beta^2) = v^2(\gamma^2 + 2\beta\delta + 2\alpha\delta),$$

$$u^4(\gamma^2 + 2\beta\delta + 2\alpha\delta) = v^2(2\alpha\gamma + 2\beta\gamma + 2\alpha\delta + \beta^2),$$

which, substituting therein for α , β , γ , δ the foregoing values, and then two equations; from these, eliminating M , we should obtain the modular equation, and then M in terms of u , v .

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Substituting in the first instance for α , δ their values, the equations are

$$u^2(2\beta + 2\gamma\beta) = v^2\left\{\gamma^2 + 2\frac{u^2}{v}(\beta + \gamma)\right\},$$

$$\gamma^2 + 2\beta\gamma + (2 + 2\beta)\frac{u^2}{v} = u^2v^2\left\{2\beta + 2\beta\gamma + 2\frac{u^2}{v} + \beta^2\right\}.$$

The first of these is

$$4(1 - uv)(2\beta + 2\gamma) + 4\frac{u^2}{v}\gamma^2 = 0,$$

viz. this is

$$4(1 - uv)\left(\frac{1}{M} - 1 + \frac{u^2}{M} - \frac{u^8}{v^3}\right) - 4\frac{u^4}{v^2} = 0,$$

or observing that the sum of the second terms is the sign of the foregoing value of $\frac{1}{M}$; so that the identity verified, it follows that the correct sign has been attributed to the value of $\frac{1}{M}$.

46. Multiplying by v , the equation is

$$-7(1 - u^2)(1 - v^2) + 7(1 - uv)(1 - v^2) - 14uv(1 - uv)^2(1 + uv) + v^2(1 - uv)^2 = 0,$$

$$+(1 - u^2)(1 - v^2) - 8(1 - uv)(1 - v^2) + 5(1 - uv)^2 = 0,$$

$$-10(4\alpha + 5u^2v - 5u^4v^2 - 5u^6v^3 - u^4) = 0,$$

or, say

$$\frac{1}{2}F'v \cdot \frac{1}{M} = 5[v^2 + 4uv^4 + 6v^2u^4 + 4u^6v + vu^2(1 - u^2)],$$

where

$$\frac{1}{2}F'v \cdot \frac{1}{M} = 3v^4 + 10vu^4 + 10v^4u^2 - 5vu^2 - 2u.$$

Hence also, reducing by the modular equation,

$$\frac{1}{2}vF'v \cdot \frac{1}{M} = 5v[v^4 + 4uv^5 + 6u^4v^2 + 2v(1 + u^2) + u^2],$$

the one of which forms is as convenient as the other.

47. Making the change u , v , $\frac{1}{M}$ into v , $-u$, $-5M$, we have

$$-\frac{1}{2}F'u \cdot 5M = 5[-u^4 + 4vu^4 + 4vu^4 + 1 - v^2(1 - v^2)];$$

and comparing with the equation

$$\frac{1}{2}vF'v \cdot \frac{1}{M} = 5[v^2 + 4u^4v + 6u^2v^4 + 4uv^5 + u^2(1 - u^2)],$$

we obtain

$$\frac{v(1-v^2)}{u(1-u^2)} = -\frac{2v(1-uv)(1-uv+u^2v^2)-vF'v}{2u(1-uv)(1-uv+u^2v^2)+uF'u}.$$

Writing for a moment $M = u^4 + 5u^2v + v^2$, $N = u^2v^4 + 5uv + v^2$, this is

$$-4Mu(1 - u^2)(1 - v^2) - (u^2 - v^2)(1 - u^2 - uv^2 - u^2v^2 + u^2v^2 + v^2u^2 + v^4) \cdot N$$

$$+ 4uN^2v^4 + (uv^4 - v^2)^2[1 - u^2v^4(uv - u^2v^2 + v^2)] = 0.$$

But we have

$$u^2(1 - uv)^2 - v^2(1 - uv)^2 = (uv - u^2v^2 - u^2v^2)[1 - u^2v^4(u^2 - v^2) + u^2v^2],$$

$$u^2(1 - v^2) + v^2(1 - u^2) = (u^2 + v^2)[1 - u^2v^4(u^2 + v^2) + u^2v^2].$$

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viz. this is

$$\begin{aligned} -7(1-v^4)(1-u^2) + 7(1-uv)(1-v^2) - 14uv(1-uv)^2(1+uv) + v^2(1-uv)^2 &= 0, \\ + (1-u^2)(1-v^2) - 8(1-uv)(1-v^2) + 5(1-uv)^2(1-uv)^2 &= 0. \end{aligned}$$

In the second column the coefficient of $2-uv-v^2$, this is

$$= (1-u^2)(1-v^2) \{[(uv)^2 - (uv)^2]\} - 14uv(1-uv)^2 - (1-uv)^2(1-uv) = 0,$$

Reducing also the other two columns by means of the modular equation, the equation thus becomes

$$\begin{aligned} -6(1-uv)^2 - (1-uv) [(1-uv)^2 - (uv)^2] - 14uv(1-uv)^2(1+uv) \\ + 8(1-uv)^2 \\ - 28uv(1-uv)^2(1-uv+u^2v^2) = 0. \end{aligned}$$

This is in fact an identity; to show it, writing for convenience θ in place of uv , and observing that the terms

$$-(1-\theta) + 8(1-\theta) + 3(1-\theta), = 5, = (1-\theta)^2 [8 - 2(1-\theta) + 5] = (1-\theta) [6 - 5\theta + 4\theta^2 - 3\theta^3]$$

the whole equation divides by $(1-\theta)^2$; or throwing out this factor, it will be observed that the terms

$$2\theta(uv - 2u^2v^2 + 2u - u^2v) = -\frac{1}{2}\theta(v^2 + 2v^2u^2 - 2vu - u^2);$$

or, say

$$\frac{1-y}{1+y} = \frac{1-x}{1+x} \left(\frac{v-\beta x+\gamma x^2-\delta x^3}{v+\beta x+\gamma x^2+\delta x^3} \right)^2,$$

but the resulting form may admit of simplification.

48. The endecadic transformation, $n = 11$.

I have not completed the solution, but the results, so far as I have obtained them, are interesting. The coefficients are $\alpha, \beta, \gamma, \delta, \epsilon, \zeta$; and we have, as in general,

$$\alpha = 1, \quad 2\epsilon = v^2u \left(\frac{1}{M} - \frac{u^{10}}{v^2} \right),$$

$$2\beta = \frac{1}{M} - 1, \quad \zeta = \frac{u^2}{v}.$$

The unused equations then are, the results, so far as I have obtained them, are interesting. The coefficients are $\alpha, \beta, \gamma, \delta, \epsilon, \zeta$; and we have, as in general,

$$u^2(2\alpha\gamma + 2\beta + \beta^2) = v^2(\gamma + 2\beta\zeta + 2\alpha\epsilon), \quad +u^4 \cdot 2\beta = 0,$$

$$u^{-2}u^4 = 2\beta \cdot \frac{u^4}{v^2} = 0,$$

$$+ \left\{ \frac{1}{u^4v^2} - \frac{u^4}{v^2} \right\} \cdot 2\gamma + \frac{1}{v^2} \left(\frac{1}{u^2v^2} - u^2v^2 \right) \cdot 2\delta = 0;$$

say, for a moment, these are

$$A + P \cdot 2\beta = 0, B + R \cdot 2\beta = 0, 1 : 2\gamma = 2\beta = PS - QR : RA - PB.$$

Here

$$\begin{aligned} PS - QR &= \frac{8\beta}{v^2}(QB - SA + \Pi), \\ &= \frac{8\beta}{v^2} \left[\frac{2u^4}{v} + (u^4v^2 - v^2) + 2u - 2u^2 \right], \\ &= -\frac{8\beta}{v^2} \left(u^4v \frac{1}{M} - \frac{u^4}{v} \right), \end{aligned}$$

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47. The conclusion is

$$\alpha = 1, \quad \beta = \frac{1}{2} \left(\frac{1}{M} - 1 \right), \quad \gamma = v^2 u \left(\frac{1}{M} - \frac{u^4}{v^2} \right), \quad \delta = \frac{u^2}{v},$$

where

$$\frac{1}{M} = \frac{-7u(1-uv)(1-uv+u^2v^2)}{u-v^2},$$

and of course

$$\frac{1-y}{1+y} = \frac{1-x}{1+x} \left(\frac{1-\beta x + \gamma x^2 - \delta x^3}{1+\beta x + \gamma x^2 + \delta x^3} \right)^2,$$

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I have not completed the solution, but the results, so far as I have obtained them, are interesting. The coefficients are $\alpha, \beta, \gamma, \delta, \epsilon, \zeta$; and we have, as in general,

$$\alpha = 1, \quad 2\epsilon = v^2 u \left(\frac{1}{M} - \frac{u^{10}}{v^2} \right),$$

$$2\beta = \frac{1}{M} - 1, \quad \zeta = \frac{u^2}{v}.$$

The unused equations then are

$$2\alpha\gamma + 2\alpha\beta + \beta^2 = \frac{u^2}{v^2} (2\alpha\gamma + 2\beta\gamma + \beta^2),$$

$$v^2 (\zeta^2 + 2\beta\zeta + 2\alpha\zeta) = u^{10} (2\alpha\gamma + 2\beta\gamma + 2\alpha\delta + \beta^2),$$

$$u^2 (\zeta^2 + 2\delta\zeta + 2\beta\zeta) + \frac{1}{v^2} \left(\frac{1}{u^2 v^2} - u^2 v^2 \right) \cdot 2\delta = 0;$$

say, for a moment, these are

$$A + P \cdot 2\beta = 0, \quad B + R \cdot 2\beta = 0, \quad 1 : 2\gamma : 2\delta = PS - QR : RA - PB : \dots$$

[The remainder of Art. 48 carries forward the elimination, expressing the coefficients γ, δ rationally in terms of $u, v, \frac{1}{M}$, and obtaining (apart from a verification which has not been carried through) the endecadic transformation in the form $\frac{1-y}{1+y} = \frac{1-x}{1+x} \left(\frac{1-\beta x + \gamma x^2 - \delta x^3 + \epsilon x^4 - \zeta x^5}{1+\beta x + \gamma x^2 + \delta x^3 + \epsilon x^4 + \zeta x^5} \right)^2$.]

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where the terms containing $\frac{1}{M}$ disappear of themselves, viz. this is

$$= \frac{4}{v} \left(\frac{u^4 v}{v^2} - 2u^2 v^4 + 2uv - u^2 v^2 \right),$$

$$= -\frac{4}{v} u^4 (v^4 + 2v^4 u^2 - 2vu - u^2);$$

observe that the term in () is equal to zero, the modular equation for the case $n = 3$. It thus appears that γ and δ are given as fractions, having in their denominator this function $u^4 + 2u^2 v - 2u^2 v^4 - v^4$.

49. To complete the calculation, we have

$$QB - AS = -uv^2 \left(\frac{v^2}{u^2 v^2} - \frac{u^4}{v^2} \beta^2 \right)$$

$$- \left[u^2 v^2 (2 + \beta) + \frac{u^4}{v^2} \left(\epsilon + 2 \frac{u^4}{v^2} \right) \right] \left\{ \frac{12}{vu^2} - \frac{u^4}{v} + \frac{\epsilon}{v^2} \right\};$$

viz. multiplying by 8, and substituting for 2β its value, this is

$$8(QB - AS) = 2uv^2 \left[v^2 u^2 \left(\frac{1}{M} - 2u^4 + 2u^2 v^2 - 1 \right) \right] \left\{ \frac{12}{u^2 v} - \frac{u^4}{v^2} - \frac{\epsilon}{v} \right\};$$

$$- \left\{ v^4 u^2 \left(\frac{1}{M} - 1 \right) - uv^2 \frac{1}{M} - \frac{u^4}{v^2} + 3 \right\} \left\{ \frac{12}{vu^2} - \frac{u^4}{v} + \frac{\epsilon}{v^2} \right\};$$

or, what is the same thing,

$$8(QB - AS) = 2u^4 \left(\frac{v^2}{u^4 v^4} - \frac{u^4}{v^4} \beta^2 \right) \\ - \left\{ v^4 u^2 \left(\frac{1}{M} - 1 \right) + \left(\frac{\beta}{M} - 1 \right) \left(\frac{1}{M} + 3 \right) \right\} \left\{ \frac{12}{vu} - \frac{u^4}{v} + \frac{\epsilon}{v^2} \right\},$$

viz. the left-hand side is

$$= 2 \{ v^2(1 - u^2 v^2) + 2v^2(1 - u^2) + u^{10} + v^4 - u \}.$$

or, say we have $-\frac{8\beta}{v^2}(QB - SA)\Pi$, where

$$\Pi = \frac{1}{Mv^2} v^2 (1 - v^2) \\ + \frac{1}{Mv^2} u^2 (u - 2u^2)(1 - v^2) \\ + \frac{1}{M} 4v^2 u^2 + v^2(1 - 3u^2) - 4uv^3 + 2u^2 \\ + \dots - 2u^4 + 6v^2 u^2(1 - v^2) + u^4(-3 + 3u^2);$$

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wherefore the value of 2γ is $= \frac{1}{u^2} v^2 (1 - v^2)$. Similarly, writing

$$\Pi' = \frac{1}{M^2} v^2 (1 - v^2) \\ + \frac{1}{M^2} u^2 (u - 2u^2)(1 - v^2) \\ + \frac{1}{M} 4v^4 u^2 + v^4(3 - v^2) + 4uv^3 + 4uv^2 \\ + v^2(-3 + 3u^2) + v^4(1 - u^2) + 4uv - 2u^4,$$

we find

$$2\delta = \frac{1}{u} \Pi' \cdot (v^2 + 2v^2 u^2 - 2vu - u^4);$$

in verification whereof observe that this being so, the first equation gives the identity

$$\left(\frac{1}{M} - 1 \right) \left(\frac{1}{M} + 3 \right) - v^2 \left(\frac{1}{M} - \frac{u^4}{v^2} \right) \left(\frac{1}{M} + \frac{2u^2 v}{v^2} \right) + \Pi - \Pi' = 0.$$

50. The result is that, writing for the moment $v^4 + 2v^2 u^2 - 2vu - u^4 = \Delta$, the values of the coefficients are

$$\alpha, \quad \beta, \quad \gamma, \quad \delta, \quad \epsilon, \quad \zeta, \\ = 1, \quad \frac{1}{2} \left(\frac{1}{M} - 1 \right), \quad \frac{1}{2\Delta} \Pi, \quad \frac{u^2}{v} \frac{\Pi'}{\Delta}, \quad \frac{1}{2} u^2 v \left(\frac{1}{M} - \frac{u^{10}}{v^2} \right), \quad \frac{u^2}{v},$$

and

$$\frac{1-y}{1+y} = \frac{1-x}{1+x} \cdot \left(\frac{1-\beta x + \gamma x^2 - \delta x^3 + \epsilon x^4 - \zeta x^5}{1+\beta x + \gamma x^2 + \delta x^3 + \epsilon x^4 + \zeta x^5} \right)^2;$$

the modular equation is known, and to complete the solution we require an expression of $\frac{1}{M}$ in terms of u, v .

51. We may herein illustrate the following theorem, viz. we may simultaneously change $u, v, \frac{1}{M}, \alpha : \beta : \gamma : \delta : \epsilon : \zeta$; into $\frac{1}{u}, \frac{1}{v}, \frac{v^2}{u^2 M}, \zeta : \epsilon : \delta : \gamma : \beta : \alpha$.

Thus making the change in the equation

$$\frac{\beta}{\alpha} = \frac{1}{2} \left(\frac{1}{M} - 1 \right),$$

we have

$$\frac{\epsilon}{\zeta} = \frac{1}{2} \left(\frac{v^2}{u^2 M} - 1 \right), \quad \text{that is,} \quad \frac{1}{2} \frac{v^2}{u^2} \left(\frac{1}{M} - \frac{u^2}{v^2} \right) = \frac{1}{2} \left(\frac{v^2}{u^2 M} - 1 \right),$$

which is right.

So in the equation $\frac{\gamma}{\alpha} = \frac{1}{2} \frac{\Pi}{\Delta}$, if for a moment $(\Pi), (\Delta)$ are what Π, Δ become, the

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equation is $\frac{\delta}{\zeta} = \frac{1}{2} \frac{(\Pi)}{(\Delta)}$, that is, $\frac{1}{u^2} \frac{\Pi'}{\Delta} = \frac{1}{u^2} \frac{(\Pi)}{(\Delta)} \Pi'$; or $(\Pi) = \frac{1}{u^2} \frac{\Delta}{\Delta'} \Pi'$; but obviously $\frac{(\Delta)}{\Delta} = -\frac{1}{u^4 v^4}$; and the equation thus is $(\Pi) = -\frac{1}{u^2 v^2} \Pi'$, or say $u^2 v^2 (\Pi) = -\Pi'$; that is,

$$\begin{aligned} -\Pi' = u^2 v^2 & \left\{ \frac{v^2}{u^4 M^2} \cdot \frac{1}{u^2} \left(1 - \frac{1}{v^2}\right) \right. \\ & + \frac{v^2}{u^4 M} \cdot \frac{1}{u^2} \left(\frac{1}{u} - \frac{2}{u^2}\right) \left(1 - \frac{1}{v^2}\right) \\ & + \frac{v^2}{u^4 M} \cdot \frac{4}{u^2 v^2} + \frac{1}{v^2} \left(1 - \frac{3}{u^2}\right) - \frac{4}{uv^3} + \frac{2}{u^2} \\ & \left. + \frac{2}{v^2} + \frac{6}{v^2 u^2} \left(1 - \frac{1}{v^2}\right) + \frac{1}{u^2} \left(-3 + \frac{3}{u^2}\right) \right\}, \end{aligned}$$

which is right.

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52. I recur to the formula

$$\frac{1-y}{1+y} = \frac{1-x}{1+x} \left(\frac{\alpha - \beta x + \gamma x^2 - \dots \pm \sigma u^{2(n-1)}}{\alpha + \beta x + \gamma x^2 + \dots + \sigma u^{2(n-1)}} \right)^2,$$

and seek to express the ratios $\alpha : \beta : \dots : \sigma$ in terms of q . Writing with Jacobi

$$\omega = \frac{mK + m' i K'}{n},$$

we have in general

$$\alpha + \beta x + \gamma x^2 + \dots + \sigma x^{2(n-1)} = \alpha \left(1 + \frac{x}{\text{snc } 2\omega}\right) \left(1 + \frac{x}{\text{snc } 4\omega}\right) \dots \left(1 + \frac{x}{\text{snc } (n-1)\omega}\right),$$

(snc = sin co am; viz. $\text{snc } 2\omega = \text{cn}(K - 2\omega)$, &c.); and the values of $\alpha, \beta, \dots, \theta$ which correspond to the moduli $v_0, v_1, \dots, v_n$, or say the values $(\alpha_0, \beta_0, \dots, \theta_0), (\alpha_1, \beta_1, \dots, \theta_1), \dots, (\alpha_n, \beta_n, \dots, \theta_n)$ are obtained by giving to ω the values

$$\frac{\omega_0}{n}, \quad \frac{\omega_1}{n}, \quad \frac{\omega_2}{n}, \quad \dots, \quad \frac{\omega_n}{n}, \quad = \frac{2K}{n}, \quad \frac{2K + 4iK'}{n}, \quad \frac{4K + 4iK'}{n}, \quad \dots, \quad \frac{iK'}{n},$$

viz. the cases ω_0, ω_n correspond to Jacobi's first and second real transformations, and the others to the imaginary transformations.

I remark that $\omega = \omega_n$ gives for $\text{snc } 2p\omega$ an expression which is rational as regards q , but $\omega = \omega_n$ gives an expression involving $q^{\frac{1}{n}}$, the real n th root of q ; the other values $\omega_1, \omega_2, \dots$ give the like expressions, involving $\alpha q^{\frac{1}{n}}, \alpha^2 q^{\frac{1}{n}}, \dots$ (α an imaginary n th root of unity), the imaginary n th roots of q .

[p. 156]

53. I consider first the expression

$$\frac{1}{\text{snc } 2p\omega_n}, \quad = \frac{1}{\text{cn}(K - 2p\omega_n)}, \quad = \frac{\text{dn } 2p\omega_n}{\text{cn } 2p\omega_n}.$$

Here, writing $2p\omega_n = \frac{2pK\xi}{\pi}$ (ξ for Jacobi's x , as x is being used in a different sense), that is,

$$\xi = \frac{\pi}{2K} \cdot 2p \cdot \frac{2K}{n}, \quad = \frac{2p\pi}{n},$$

(and hence $e^{i\xi} = e^{\frac{2pi\pi}{n}} = \alpha^p$, $e^{i\xi} = \alpha^{2p}$, if $\alpha = e^{\frac{2i\pi}{n}}$, an imaginary n th root of unity), then (Jacobi, p. 86, [*Ges. Werke*, t. i., p. 143])

$$\begin{aligned} \frac{1}{\text{snc } 2p\omega_n} &= \text{dn } \frac{2K\xi}{\pi} \div \text{cn } \frac{2K\xi}{\pi} \\ &= \frac{C}{B} \cdot \frac{2q^{\frac{1}{4}}}{c^{\frac{1}{4}}} \cdot \frac{(1 - q^{\alpha p^2}) \dots (1 + q^{\alpha p^2}) \dots}{(1 + q^{\alpha p^2}) \dots (1 + q^{\alpha p^2}) \dots}, \end{aligned}$$

where

$$\frac{C}{B} = \frac{[(1+q^2)\dots]^2}{[(1+q)\dots]^2}, = f^2(q);$$

that is,

$$\frac{1}{\operatorname{snc} 2p\omega_n} = \frac{2qi^2}{\pi} f^2(q) \frac{(1+\alpha^{2p}q)\dots(1+\alpha^{-2p}q)\dots}{(1+\alpha^{2p}q^2)\dots(1+\alpha^{-2p}q^2)\dots},$$

where, for shortness, I write $(1+q^{\alpha p^2})\dots$ to denote the infinite product

$$(1+q^{\alpha p^2})(1+q^{\alpha p^2})(1+q^{\alpha p^2})\dots,$$

and similarly $(1+q^2q^{\alpha p^2})\dots$ to denote the infinite product $(1+q^2q^{\alpha p^2})(1+q^4q^{\alpha p^2})(1+q^6q^{\alpha p^2})\dots$, and the like for the terms in $e^{-i\xi}$; the notation, accompanied by its explanation, is quite intelligible, and it would be difficult to make one which would be at the same time complete and not cumbrous. Then attributing to g the values $1, 2, \dots, \frac{1}{2}(n-1)$, and forming the symmetric functions of these expressions, we have the values of $\frac{\beta}{\alpha}, \frac{\gamma}{\alpha}$, &c, or α being put $= 1$, say the values of $\beta, \gamma, \dots, \sigma$.

54. I stop to notice a verification afforded by the value of β_n . Putting $u = 0$, that is, $q = 0$, we have

$$\frac{1}{\operatorname{snc} 2p\omega_n} = \frac{2\pi}{1+\alpha^{2p}},$$

and thence

$$\beta_n = 2\left\{\frac{1}{1+\alpha^2} + \frac{1}{1+\alpha^4} + \frac{1}{1+\alpha^6} + \dots + \frac{1}{1+\alpha^{n-1}}\right\};$$

we have $2\beta_n = \frac{1}{M_n} - 1$; and putting as above $u = 0$, the value of $\frac{1}{M_n}$ is $(-)^{\frac{n-1}{2}} n$; whence

$$(-)^{\frac{n-1}{2}} n - 1 = 4\left\{\frac{\alpha}{1+\alpha^2} + \frac{\alpha^2}{1+\alpha^4} + \frac{\alpha^3}{1+\alpha^6} + \dots + \frac{\alpha^{(n-1)/2}}{1+\alpha^{n-1}}\right\},$$

[p. 157]

a theorem relating to the imaginary n th roots of unity, α an odd prime. In particular,

$$n = 3, \quad -4 = 4\left\{\frac{\alpha}{1+\alpha^2}\right\}, \text{ at once verified by } \alpha^2 + \alpha + 1 = 0;$$

$$n = 5, \quad 4 = 4\left\{\frac{\alpha}{1+\alpha^2} + \frac{\alpha^2}{1+\alpha^4}\right\}, \text{ verified by } \alpha^4 - 1 = 0,$$

viz. the theorem is also true for the real root $\alpha = 1$; in fact, the term in [] is $\{\alpha(1+\alpha^4) + \alpha^2(1+\alpha^2)\} \div (1+\alpha^2)(1+\alpha^4)$, $= (\alpha + 1 + \alpha^4 + \alpha^4) \div (1 + \alpha^2 + \alpha^4 + \alpha^6)$, $= 1$;

$$n = 7, \quad -8 = 4\left\{\frac{\alpha}{1+\alpha^2} + \frac{\alpha^2}{1+\alpha^4} + \frac{\alpha^3}{1+\alpha^6}\right\},$$

which may be verified by means of $\alpha^7 - \alpha^6 + \alpha^5 - \alpha^4 + \alpha^3 - \alpha^2 + \alpha + 1 = 0$; and so on.

55. I further remark that we have

$$\frac{1}{M_n} = (-)^{\frac{n-1}{2}} \left\{ \frac{\operatorname{sn} 2\omega_n \operatorname{sn} 4\omega_n \dots \operatorname{sn}(n-1)\omega_n}{\operatorname{snc} 2\omega_n \operatorname{snc} 4\omega_n \dots \operatorname{snc}(n-1)\omega_n} \right\}^2.$$

But Jacobi (p. 86, [l.c.]),

$$\operatorname{sn} 2p\omega_n = \frac{2K\xi}{\pi}, \quad = \frac{AK}{\pi i^4} \frac{e^{i\xi} - 1}{e^{i\xi}} \frac{(1-q^{\alpha p^2})\dots(1-q^{-\alpha p^2})\dots}{(1-q^{\alpha}q^{\alpha p^2})\dots(1-q^{\alpha}q^{-\alpha p^2})\dots},$$

where (p. 89, [l.c., p. 146])

$$\frac{AK}{\pi} = \frac{\sqrt{q}}{\sqrt{k}}, \quad = \frac{1}{2} \frac{[(1+q)(1+q^2)\dots]^2}{[(1+q^2)\dots(1+q^4)\dots]^2}, \quad = \frac{1}{2} f^{-2}(q),$$

that is,

$$\operatorname{sn} 2p\omega_n = f^{-2} q^{\frac{n-1}{4}} \cdot \frac{(1-q^{\alpha p^2})\dots(1-q^{-\alpha p^2})\dots}{(1-q^{\alpha}q^{\alpha p^2})\dots(1-q^{\alpha}q^{-\alpha p^2})\dots}.$$

Hence

$$\frac{\operatorname{sn} 2p\omega_n}{\operatorname{snc} 2p\omega_n} = \frac{q^n-1}{i(q^n+1)} \cdot \frac{1-q^{\alpha p^2}}{1+q^{\alpha p^2}} \cdot \frac{(1+q^{\alpha p^2})\dots(1+q^{-\alpha p^2})\dots}{(1-q^q q^{\alpha p^2})\dots(1-q^q q^{-\alpha p^2})\dots} \cdot \frac{(1+q^q q^{\alpha p^2})\dots(1+q^q q^{-\alpha p^2})\dots}{(1-q^{\alpha p^2})\dots(1-q^{-\alpha p^2})\dots},$$

and giving to g the values $1, 2, \dots, \frac{1}{2}(n-1)$, and multiplying the several expressions, we have the value of $\frac{1}{M_n}$, viz. observing that we have in the denominator $(i)^{\frac{1}{2}(n-1)} = (-)^{\frac{n-1}{2}}$ which destroys this factor in the expression of $\frac{1}{M_n}$, this is

$$\frac{1}{M_n} = \Pi \left\{ \frac{1-q^{\alpha p^2}}{1+q^{\alpha p^2}} \cdot \frac{(1+q^{\alpha p^2})\dots(1+q^{-\alpha p^2})\dots}{(1-q^q q^{\alpha p^2})\dots(1-q^q q^{-\alpha p^2})\dots} \right\}^4,$$

[p. 158]

In fact, $n = 3$, this is

$$\frac{(\alpha^2-1)^2}{(\alpha^2+1)^2} = -3,$$

viz. the numerator is $\alpha - 2\alpha^2 + 1, = -3\alpha^2$, and the denominator is $(-\alpha)^2, = \alpha^2$.

So $n = 5$, the formula is

$$\left(\frac{\alpha^2-1}{\alpha^2+1} \frac{\alpha^4-1}{\alpha^4+1} \right)^2 = 5, \text{ that is, } \left(\frac{(\alpha^2-1)^2}{(\alpha^4+1)^2} \right)^2 = 5,$$

or

$$\frac{\alpha^4-4\alpha^3+6\alpha^2-4\alpha+1}{\alpha^8+2\alpha^4+1} = 5,$$

viz. this is $5(1 + \alpha + 2\alpha^2) - (1 - 4\alpha + 6\alpha^2 - 4\alpha^2 + 6\alpha^4) = 0$, which is right; and so in other cases.

We thus have

$$\frac{1}{M_n} = (-)^{\frac{n-1}{2}} n \cdot R(q),$$

which, on putting therein $u = 0$, that is, $q = 0$, gives, as it should do, $\frac{1}{M_n} = (-)^{\frac{n-1}{2}} n$.

57. As regards the expression of $R(q)$, observe that, giving to g its different values, the factors $1 - \alpha^{2p}q$ and $1 - \alpha^{-2p}q$ are the factors other than $1 - q$ of $1 - q^n$, and so as to the other pairs of factors; viz. we have

$$R(q) = \left(\frac{1-q^n}{1-q} \frac{1+q^2}{1+q^{2n}} \frac{1+q}{1+q^n} \right)^4,$$

viz. this is

$$= \left(\frac{1-q^n}{1+q^n} \frac{1+q^2}{1-q^2} \right)^4, = \left(\frac{1-q^n}{1+q^n} \frac{1+q}{1-q} \right)^4,$$

that is,

$$\frac{1}{M_n} = (-)^{\frac{n-1}{2}} n \frac{\phi^4(q^n)}{\phi^4(q)},$$

agreeing with a former result.

58. We have of course the identity $2\beta_n = \frac{1}{M_n} - 1$; that is,

$$4 \sum \frac{e^{i\xi}}{1+q e^{i\xi}} f^2(q) \frac{(1+q^q q^{\alpha p^2})\dots(1+q^q q^{-\alpha p^2})\dots}{(1+q^{\alpha p^2})\dots(1+q^{-\alpha p^2})\dots} = (-)^{\frac{n-1}{2}} n \frac{\phi^4(q^n)}{\phi^4(q)} - 1,$$

($g = 1, 2, \dots, \frac{1}{2}(n-1)$), which, putting therein $q = 0$, is an identity before referred to; a form perhaps more convenient is obtained by dividing each side by $f^2(q)$.

59. I notice further that we have

$$v_0 = u^4 \{ \operatorname{snc} 2\omega_n \operatorname{snc} 4\omega_n \dots \operatorname{snc}(n-1)\omega_n \};$$

the term in [] is

$$\Pi \frac{1+\alpha^{2p}}{2q^{\frac{1}{4}}} f^{-2}(q) \frac{(1+q^{\alpha p^2})\dots(1+q^{-\alpha p^2})\dots}{(1+q^q q^{\alpha p^2})\dots(1+q^q q^{-\alpha p^2})\dots},$$

where we have $\Pi \frac{1+\alpha^{2p}}{2} = (-)^{\frac{n-1}{2}}$. For example, $n = 3$, the term is $\frac{1+\alpha^2}{2} = -1$;

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$n = 5$, it is

$$\frac{(1+\alpha^2)(1+\alpha^4)}{\alpha \cdot \alpha^2 \cdot \alpha^4}, = \frac{1+\alpha^4+\alpha^2+\alpha^4}{\alpha^7}, = -1;$$

$n = 7$, it is

$$\frac{(1+\alpha^2)(1+\alpha^4)(1+\alpha^6)}{\alpha \cdot \alpha^2 \cdot \alpha^4 \cdot \alpha^4}, = \frac{1+\alpha+\alpha^4+\alpha^4+\alpha^4+\alpha^4+2\alpha^7}{\alpha^7} = 1;$$

and so on. The term in question thus is

$$(-)^{\frac{n-1}{2}} \frac{1}{(\sqrt{2})^{n-1}} f^{-2n}(q) \frac{1+q^{\alpha p^2} \dots 1+q \dots}{1+q^q q^{\alpha p^2} \dots 1+q^{\frac{q^2}{2}} \dots}.$$

This has to be multiplied by $u^4, = (\sqrt{2})^4 q^{\frac{1}{2}} f^2(q)$, and we thus obtain

$$v_0 = (-)^{\frac{n-1}{2}} \sqrt{2} q^{\frac{1}{2}} f^4(q^n),$$

agreeing with a former result.

We have in regard to the value of $\frac{1}{M_n}$ a verification afforded by the value of M_n ; then, as well the new modulus $\lambda, (= v_0^4)$ and the multiplier M_n , as also the several functions which enter into the expression

$$\frac{1-y}{1+y} = \frac{1-x}{1+x} \left(\frac{\left(1 - \frac{x}{\text{snc } 2\omega_n}\right) \dots \left(1 - \frac{x}{\text{snc } (n-1)\omega_n}\right)}{\left(1 + \frac{x}{\text{snc } 2\omega_n}\right) \dots \left(1 + \frac{x}{\text{snc } (n-1)\omega_n}\right)} \right)^2,$$

are all of them expressed as functions of q .

60. I consider in like manner the expression

$$\frac{1}{\text{snc } 2p\omega_n}, = \frac{1}{\text{sn}(K-2p\omega_n)}, = \frac{\text{dn } 2p\omega_n}{\text{cn } 2p\omega_n}.$$

Here, writing $2p\omega_n = \frac{2pK\xi}{\pi}$ (ξ instead of Jacobi's x as before), that is,

$$\xi = \frac{\pi}{2K} \cdot 2p \cdot \frac{iK'}{n}, = \frac{p\pi iK'}{nK},$$

and thence

$$e^{i\xi} = e^{-\frac{p\pi K'}{nK}}, = q^{\frac{p}{n}},$$

we have

$$\begin{aligned} \frac{1}{\text{snc } 2p\omega_n} &= \text{dn } \frac{2K\xi}{\pi} \div \text{cn } \frac{2K\xi}{\pi} \\ &= f^2(q) \frac{2q^{\frac{1}{4}}}{\pi} \frac{(1+q^{\frac{2p}{n}})(1+q^{2-\frac{2p}{n}}) \dots (1+q^{1-\frac{2p}{n}}) \dots}{1+q^{\frac{p}{n}}(1+q^{1+\frac{p}{n}})(1+q^{1-\frac{p}{n}}) \dots (1+q^{1-\frac{p}{n}}) \dots}. \end{aligned}$$

[p. 160]

where the notations are as follows:

$$(1 + q^{1+\frac{2p}{n}}) \dots \text{ is the infinite product } (1 + q^{1+\frac{2p}{n}})(1 + q^{2+\frac{2p}{n}})(1 + q^{3+\frac{2p}{n}}) \dots,$$

and

$$(1 + q^{1+\frac{2p}{n}}) \dots \text{ is the infinite product } (1 + q^{1+\frac{2p}{n}})(1 + q^{2+\frac{2p}{n}})(1 + q^{3+\frac{2p}{n}}) \dots;$$

and the like as to the expressions with exponents containing $-\frac{2p}{n}$.

And then attributing to g the values $1, 2, \dots, \frac{1}{2}(n-1)$, and forming the symmetric functions of these expressions, we have the values of $\frac{\beta}{\alpha}, \frac{\gamma}{\alpha}$, &c, or α being put $= 1$, say the values of $\beta, \gamma, \dots, \sigma$.

It is easy to see, and I do not stop to prove that, if instead of $\omega = \omega_n$ we have $\omega = \omega_1, \omega_2, \dots$, or ω_{n-1} , we simply multiply $q^{\frac{1}{n}}$ by an imaginary n th root of unity; that is, we replace the real n th root $q^{\frac{1}{n}}$ by an imaginary n th root of q .

In the case $u = 0$, that is, $q = 0$, do not stop to prove here, if instead of $\omega = \omega_n$ we have $\omega = \omega_1, \omega_2, \dots, \omega_{n-1}$, we simply multiply $q^{\frac{1}{n}}$ by an imaginary n th root of unity; that is, we replace the real n th root $q^{\frac{1}{n}}$ by an imaginary n th root of q .

In the case $u = 0$, that is, $q = 0$, the modulus $\lambda, (= v_0^4), = 0$; and the like for the values $\omega_1, \omega_2, \dots, \omega_{n-1}$, the equation $2\beta = \frac{1}{M} - 1$ gives consequently for $\frac{1}{M}$, n values each $= 1$, agreeing with the multiplier equation.

61. We have for M_n the formula

$$\frac{1}{M_n} = (-)^{\frac{n-1}{2}} \left\{ \frac{\text{sn } 2\omega_n \text{ sn } 4\omega_n \dots \text{sn}(n-1)\omega_n}{\text{snc } 2\omega_n \text{ snc } 4\omega_n \dots \text{snc}(n-1)\omega_n} \right\}^2,$$

and, as before,

$$\text{sn } 2p\omega_n = f^{-2}(q) \frac{q^{\frac{1}{n}} - 1}{(1 - q^{\frac{2p}{n}})(1 - q^{2 - \frac{2p}{n}}) \dots (1 - q^{1 - \frac{2p}{n}}) \dots} 2iq^{\frac{1}{4}} \cdot \frac{(1 + q^{\frac{p}{n}})(1 + q^{1 + \frac{p}{n}}) \dots (1 + q^{1 - \frac{p}{n}}) \dots}{(1 + q^{1 + \frac{p}{n}})(1 + q^{1 - \frac{p}{n}}) \dots (1 + q^{1 - \frac{p}{n}}) \dots},$$

hence

$$\frac{\text{sn } 2p\omega_n}{\text{snc } 2p\omega_n} = \frac{q^{\frac{p}{n}} - 1}{i(q^{\frac{p}{n}} + 1)} \cdot \frac{1 - q^{\frac{p}{n}}}{1 + q^{\frac{p}{n}}} \cdot \frac{(1 + q^{\frac{p}{n}}) \dots (1 + q^{1 - \frac{p}{n}}) \dots}{(1 - q^{1 + \frac{p}{n}}) \dots (1 - q^{1 - \frac{p}{n}}) \dots} \cdot \frac{(1 + q^{1 + \frac{p}{n}}) \dots (1 + q^{1 - \frac{p}{n}}) \dots}{(1 - q^{\frac{p}{n}}) \dots (1 - q^{1 - \frac{p}{n}}) \dots},$$

and we thence derive the value of $\frac{1}{M_n}$, viz. observing that we have in the denominator $(i)^{\frac{1}{2}(n-1)}, = (-)^{\frac{n-1}{2}}$

which destroys this factor in the expression of $\frac{1}{M_n}$, this is

$$\frac{1}{M_n} = \Pi \left\{ \frac{1 - q^{\frac{p}{n}}}{1 + q^{\frac{p}{n}}} \cdot \frac{(1 + q^{\frac{p}{n}}) \dots (1 + q^{1 - \frac{p}{n}}) \dots}{(1 - q^{1 + \frac{p}{n}}) \dots (1 - q^{1 - \frac{p}{n}}) \dots} \cdot \frac{(1 + q^{1 + \frac{p}{n}}) \dots (1 + q^{1 - \frac{p}{n}}) \dots}{(1 - q^{\frac{p}{n}}) \dots (1 - q^{1 - \frac{p}{n}}) \dots} \right\}^4.$$

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Now, giving to g its values, it is easy to see that we have

$$\Pi (1 - q^{\frac{2p}{n}})(1 - q^{2 - \frac{2p}{n}}) \dots (1 - q^{2 - \frac{2p}{n}}) \dots = \frac{(1 - q^{\frac{1}{n}}) \dots}{(1 - q^n) \dots},$$

where $(1 - q^{\frac{1}{n}}) \dots$ denotes $(1 - q^{\frac{1}{n}})(1 - q^{\frac{2}{n}})(1 - q^{\frac{3}{n}}) \dots$, viz. it is the same function of $q^{\frac{1}{n}}$ that $(1 - q^n) \dots$ is of q ; also

$$\Pi (1 + q^{\frac{2p}{n}})(1 + q^{2 - \frac{2p}{n}})(1 + q^{2 - \frac{2p}{n}}) \dots = \frac{(1 + q^{\frac{1}{n}}) \dots}{(1 + q) \dots},$$

where $(1 + q^{\frac{1}{n}}) \dots$ denotes $(1 + q^{\frac{1}{n}})(1 + q^{\frac{2}{n}})(1 + q^{\frac{3}{n}}) \dots$, viz. is the same function of $q^{\frac{1}{n}}$ that $(1 + q) \dots$ is of q ; and the like as to the denominator factors; we thus have

$$\frac{1}{M_n} = \left\{ \frac{(1 - q^{\frac{1}{n}}) \dots (1 + q^{\frac{1}{n}}) \dots (1 + q^{\frac{1}{n}}) \dots (1 - q) \dots}{(1 - q^n) \dots (1 + q^n) \dots (1 + q) \dots (1 - q^{\frac{1}{n}}) \dots} \right\}^4,$$

viz. this is

$$= \left\{ \frac{(1 - q^{\frac{1}{n}}) \dots (1 + q^{\frac{1}{n}}) \dots}{(1 + q^n) \dots (1 - q^n) \dots} \right\}^4 \div \left\{ \frac{(1 + q^n) \dots (1 + q) \dots}{(1 + q^n) \dots (1 - q) \dots} \right\}^2;$$

or, we have

$$\frac{1}{M_n} = \phi^4(q^{\frac{1}{n}}) \div \phi^4(q),$$

agreeing with a former result.

We have

$$2\beta_n = \frac{1}{M_n} - 1,$$

that is,

$$2 \left\{ \frac{1}{\text{snc } 2\omega_n} + \frac{1}{\text{snc } 4\omega_n} + \dots + \frac{1}{\text{snc}(n-1)\omega_n} \right\} = \frac{\phi^4(q^{\frac{1}{n}})}{\phi^4(q)} - 1,$$

a result which, substituting on the left-hand side the foregoing values of the several functions, must be identically true.

62. We have also

$$v_n = u^4 \{ \text{snc } 2\omega_n \text{ snc } 4\omega_n \dots \text{snc}(n-1)\omega_n \},$$

where the term in [] is

$$= \Pi f^{-2}(q) \frac{(1+q^{\frac{2p}{n}})(1+q^{1-\frac{2p}{n}})\dots(1+q^{1+\frac{2p}{n}})\dots}{2q^{\frac{1}{n}}(1+q^{1+\frac{2p}{n}})\dots(1+q^{1-\frac{2p}{n}})\dots},$$

or, observing that the sum of the exponents $\frac{g}{n}$ is $\frac{1}{n}[1+2\dots+\frac{1}{2}(n-1)] = \frac{n^2-1}{8n}$, this is

$$= f^{-n+1}(q) \frac{1}{(\sqrt{2})^{n-1} q^{(n^2-1)/(8n)}} \frac{(1+q^{2/n})\dots(1+q)\dots}{(1+q^{1+2/n})\dots(1+q)\dots};$$

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or, the last factor being $f(q^{\frac{1}{n}}) \div f(q)$, the expression is

$$f^{-n}(q) \frac{1}{(\sqrt{2})^{n-1}} q^{-\frac{n^2-1}{8n}} f(q^{\frac{1}{n}});$$

or, multiplying by $u^4 = (\sqrt{2})^4 q^{\frac{1}{2}} f^2(q)$, we have

$$v_n = \sqrt{2} q^{\frac{1}{n} - \frac{n^2-1}{8n}} f^4(q^{\frac{1}{n}}),$$

agreeing with a former result.

We have in like manner the complete q -transcendental solution for the *trans-formatio secunda*; viz. the original modulus $k (= u^4)$ being given as a function of q , then, as well the new modulus $\lambda, (= v_n^4)$ being also a function of M_n , as also the several functions which enter into the expression

$$\frac{1-y}{1+y} = \frac{1-x}{1+x} \left(\frac{(1-\frac{x}{\text{snc } 2\omega_n})\dots(1-\frac{x}{\text{snc}(n-1)\omega_n})}{(1+\frac{x}{\text{snc } 2\omega_n})\dots(1+\frac{x}{\text{snc}(n-1)\omega_n})} \right)^2,$$

are all expressed as functions of q .

63. As an illustration of the formulae for the *transformatio secunda* I write $n = 7$; and putting for greater convenience $q = r^7$, that is, $r = q^{\frac{1}{7}}$, then we have

$$v_7 = \sqrt{2} r^{\frac{1}{2}} f(r), \quad \frac{1}{M_7} = \phi^4(r) \div \phi^4(r^7),$$

$$\frac{1}{\text{snc } 2\omega_7} = 2f^2(r^7) A, \quad \frac{1}{\text{snc } 4\omega_7} = 2f^2(r^7) B, \quad \frac{1}{\text{snc } 6\omega_7} = 2f^2(r^7) C,$$

where

$$\begin{aligned} A &= r^4 \cdot \frac{5.19\dots 9.23\dots}{2.16\dots 12.26\dots}, \\ B &= r^4 \cdot \frac{3.17\dots 11.25\dots}{4.18\dots 10.24\dots}, \\ C &= r^4 \cdot \frac{1.15\dots 13.27\dots}{6.20\dots 8.22\dots}, \end{aligned}$$

where the numerator of A denotes $(1+r^5)(1+r^{19})\dots(1+r^9)(1+r^{23})\dots$, and so in other cases, the difference of the exponents being always = 14. And we have, as mentioned, the identical equation

$$f^4(r) (A + B + C) = \frac{1}{4} \left(\frac{\phi^4 r}{\phi^4 r^7} - 1 \right).$$

The values of the several expressions up to r^{24} are as follows: Mr J. W. L. Glaisher kindly performed for me the greater part of the calculation.

[p. 163] Table I (the index of r vs. coefficients A , B , C , Sum, $\phi^4(r) \div \phi^4(r^7)$)

[Table: a 4-column numerical table giving, for each integer power r from 0 to 50, the entries of A , B , C and their Sum, with the running value of the multiplier $\phi^4(r^7) \div \phi^4(r^7)$.]

Ind. of r	A	B	C	Sum	Mult. by $f^4(r^7) \div \phi^4(r^7)$	
0				0	+	1
1	+ 1			+ 1	—	4
2		+ 1		+ 1	+	4
3	— 1			— 1	—	0
4			+ 1	+ 1	—	0
5	+ 1		+ 1	+ 2	+	8
6	+ 1	— 1		0	—	8
7		— 1		— 1	—	0
8	— 1			— 1	—	5
9		+ 1		+ 1	—	12
10	+ 2	+ 1	— 1	+ 2	+	8
11	— 1		— 1	— 2	—	32
12	— 2		+ 1	— 1	+	8
13		+ 2	+ 1	+ 3	+	13
14	+ 1	+ 1		+ 2	+	32
15	+ 2			+ 2	+	36
16		— 2	+ 2	0	+	24
17	— 2	— 2		— 4	—	32
18	— 1		+ 2	+ 1	—	13
19	+ 2		+ 2	+ 4	+	96
20		+ 3		+ 3	—	24
21	— 2	+ 2	— 2	— 2	—	32
22	+ 1		— 1	0	—	80
23	— 3	+ 2	+ 3	+ 2	—	56
24	— 4		+ 1	— 3	+	132
25		— 4		— 4	—	248
26	+ 5	+ 2	— 3	+ 4	+	64
27	+ 3	+ 1	— 3	+ 1	—	76
28		— 1	+ 2	+ 1	+	184
29	— 5	+ 4	+ 5	+ 4	+	56
30	— 3		+ 5	+ 2	—	160
31	+ 1	+ 4		+ 5	+	77
32	+ 9	+ 5	— 4	+ 9	+	208
33	+ 4		— 5	— 1	+	576
34	— 7	— 8	+ 1	— 14	—	152
35		— 5		— 5	—	396
36	+ 7	+ 2	— 9	0	—	488
37	+ 11	+ 10	— 11	+ 10	+	332
38	+ 1		— 10	— 9	—	672
39	— 13		+ 5	— 28	+	1240
40	— 2	+ 13	+ 8	+ 8	+	328
41	+ 13		— 6	+ 4	+	352
42	+ 9	— 9	+ 5	+ 7	+	816
43	— 14	— 14	+ 14	— 14	—	1008
44	— 14	— 14	+ 16	— 12	—	728
45	— 1	— 9	+ 13	+ 3	+	1376
46	+ 17	+ 11	+ 12	+ 40	—	2328
47	+ 3	— 20	— 12	— 12	—	672
48	— 6	— 23	— 1	— 30	+	1376
49	— 17	+ 13	+ 5	+ 1	—	700
50	— 13	+ 5	— 14	— 22	—	1504

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64. As already mentioned, the foregoing expressions of the coefficients in terms of q may be applied to the determination of the coefficients as rational functions of u, v .

Representing by θ any one of the coefficients $\alpha, \beta, \gamma, \dots, \sigma$, consider the sum

$$S \frac{v^p \theta}{u^q},$$

f a positive integer, and the summation extending as before to the $n + 1$ values of v , and corresponding values of $\frac{\theta}{u}$. This is a rational function of u , and it is also integral. As to this observe that the function, if not integral, would become infinite either for $u = 0$ (this would mean that the expression contained a term or terms Au^{-m}) or for some finite value of u . But the function can only become infinite by reason of some term or terms of $Su^p \frac{\theta}{u}$ becoming infinite; viz. some term $\frac{1}{\text{snc } 2p\omega}$ must become infinite; or attending to the equation

$$v_n = u^4 \{\text{snc } 2\omega \text{ snc } 4\omega \dots \text{snc } (n-1)\omega\},$$

it can only happen if $u = 0$, or if $v = \infty$, and from the modular equation it appears that if $v = \infty$, then also $u = \infty$. Now $u = 0$ gives the ratios $\frac{\beta}{\alpha}, \frac{\gamma}{\alpha}, \dots$, each of them infinite at most in $\frac{1}{v^p}$ (this would mean that the expression contained a term or terms Au^{-m}) or for some finite value of u . Now $u = 0$ gives the values $\frac{\beta}{\alpha}, \frac{\gamma}{\alpha}, \dots$, each of them become infinite for $u = 0$ either for $u = 0$ (this would mean that the expression contained a term or terms Au^{-m}) or for some finite value of u . But the function can only become infinite by reason of some term or terms of $Su^p \frac{\theta}{u}$ becoming infinite; viz. some term $\frac{1}{\text{snc } 2p\omega}$ must become infinite; or attending to the equation

$$v_n = u^4 \{\text{snc } 2\omega \text{ snc } 4\omega \dots \text{snc } (n-1)\omega\},$$

it can only happen if $u = 0$, or if $v = \infty$, and from the modular equation it appears that if $v = \infty$, then also $u = \infty$. Now $u = 0$ gives the ratios $\frac{\beta}{\alpha}, \frac{\gamma}{\alpha}, \dots$, each of them infinite at most in $\frac{1}{v^p}$; so that there having a denominator a power of u , and if the highest powers of u in $Sv^p \theta$ is the lowest negative power (highest positive power) being $\frac{1}{u^p}$; fractional power $q^{\frac{p}{n}}$ acquires by the summation the coefficient $(1 + \alpha + \alpha^2 + \dots + \alpha^{n-1})$, and therefore disappears; there remains only the radicality $q^{\frac{p}{n}}$ occurring in the expression of the v 's; and if $nf u \pmod{8}, \mu = 0$, or a positive integer less than n , then in the form of the expression in $q^{\frac{1}{n}}$ into a rational function of q . Hence this, being a rational and integral function of u , must be of the form

$$Au^p + Bu^{p+8} + Cu^{p+16} + \&c.$$

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65. We have thus in general

$$Sv^p \frac{\theta}{u^q} = Au^p + Bu^{p+8} + \&c.;$$

and in like manner

$$Sv^p \frac{\theta}{u^q} = A'u^{p-q} + B'u^{p-q+8} + \&c.$$

We may in these expressions find a limit to the number of terms, by means of the before-mentioned theorem that we may simultaneously interchange $u, v; \alpha, \beta, \dots, \rho, \sigma$ into $\frac{1}{u}, \frac{1}{v}; \sigma, \rho, \dots, \beta, \alpha$. Starting from the expression of $Sv^p \frac{\theta}{u^q}$, let ϕ be the corresponding coefficient to θ ; viz. in the series $\alpha, \beta, \dots, \theta, \dots, \phi, \dots, \rho, \sigma$, let ϕ be as removed from σ as θ is from α ; then the equation becomes

$$Sv^{-p} \frac{\phi}{v^q} = Av^{-p} + Bv^{-p-8} + \&c.,$$

where

$$\frac{\phi}{\sigma} = \frac{\phi}{a} \cdot \frac{a}{\sigma} = \frac{\pi}{u^2} \frac{\phi}{a}; \text{ the equation thus is}$$

$$Sv^p \frac{\phi}{a} = Av^p + Bv^{p-8} + \&c.,$$

and by what precedes the series on the right-hand side can contain no negative power higher than $\frac{1}{u^{p+q+8}}$; that is, the series of coefficients $A, B, C, \dots$ goes on to a certain point only, the subsequent coefficients all of them vanishing.

In like manner from the equation $Sv^p \frac{\theta}{u^q}$ we have

$$Sv^{p+q} \frac{\phi}{a} = A'v^{n+p+q} + B'v^{n+p+q-8} + \&c.,$$

where the indices must be positive; viz. the series of coefficients $A', B', \dots$ goes on to a certain point only, the subsequent coefficients all of them vanishing.

67. The like theory applies to the expression $\frac{1}{M_n}$. We have, putting as before $uf \equiv \mu \pmod{8}$,

$$Sv^p \frac{1}{M_n} = Au^p + Du^{p+8} + \&c.,$$

$$Sv^p \frac{1}{M_n} = A'u^{p-q} + B'u^{p-q+8} + \dots$$

and we find a limit to the number of terms by the consideration that we may simultaneously change u, v , $\frac{1}{M_n}$ into $\frac{1}{u}, \frac{1}{v}, \frac{v^2}{u^2 M_n}$; the equations thus become

$$Sv^{-p} \frac{1}{M_n} = Av^{-p} + Bv^{-p-8} + \&c. \dots$$

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(where, if $f =$ or < 4 , there must be on the right-hand side no negative power of u ; but if $f > 4$, then the highest negative power must be $\frac{1}{u^{(p+q-4)/8}}$, and)

$$Sv^{p+q} \frac{1}{M_n} = A'v^{n+p+q} + B'v^{n+p+q-8} + \dots,$$

where on the right-hand side there must be no negative power of v .

68. It is to be remarked that β, ρ being always given linearly in terms of $\frac{1}{M}$, it is the same thing whether we seek in this manner for the values of β, ρ , or for that of $\frac{1}{M}$; but the latter course is practically more convenient. Thus in the cases $n = 5, n = 7$ we require only the value of $\frac{1}{M}$.

In the case $n = 11$, where the coefficients are $\alpha, \beta, \gamma, \delta, \epsilon, \zeta$, it has been seen that γ, δ are given as cubic functions of $\frac{1}{M}$; seeking for them directly, their values would (if the process be practicable) be obtained in a better form, viz. instead of the denominator $(F'v)^2$ there would be only the denominator $F'(v)$.

69. I consider for $\frac{1}{M}$ the cases $n = 3$ and 5:

$$n = 3, f = 0, 1, 2, 3, \text{ then } \mu = 0, 3, 6, 1;$$

and we write down the equations

$$S \frac{1}{M} = A, \quad \text{giving } S \frac{v^0}{M} = Au^0,$$

$$S \frac{v}{M} = A'u^3, \quad , , \quad S \frac{v^2}{M} = A'u^3,$$

$$S \frac{v^2}{M} = 0, \quad , , \quad S \frac{v^2}{M} = 0;$$

viz. if we had in the first instance assumed $S \frac{1}{M} = A + Bu^8 + \dots$, this would have given $S \frac{v^2}{M} = Au^2 + Bu^{10} + \dots$, whence B and the succeeding coefficients all vanish; and so in other cases. We have here only the coefficients A, A' ; and these can be obtained without the aid of the q -formula by the consideration that for $u = 1$ the corresponding values of $v, \frac{1}{M}$ are

$$v = 1, -1, -1, -1, -1,$$

$$\frac{1}{M} = 3, -1, -1, -1, -1.$$

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whence $A = A' = 6$; or we have the equations

$$S \frac{1}{M} = 0, \quad S \frac{v}{M} = 6u^4, \quad S \frac{v^2}{M} = 0, \quad S \frac{v^2}{M} = 6u,$$

giving as before

$$(2v^4 + 3v^2u - v) \frac{1}{M} = 3(v^2u^4 + 2u^2v + 1)u,$$

reducible by means of the modular equation to $\frac{1}{M} = 1 + \frac{2u^2}{v}$.

70. $n = 5$. Corresponding to $f = 0, 1, 2, 3, 4, 5$, we have $\mu = 0, 5, 2, 7, 4, 1$, and we find

$$\begin{array}{ll} S \frac{1}{M} = A, & \text{giving } S \frac{v^0}{M} = Au^0, \\ S \frac{v}{M} = 0, & S \frac{v^2}{M} = 0, \\ S \frac{v^2}{M} = A'u^2, & S \frac{v^2}{M} = A'u^2, \\ S \frac{v^2}{M} = A''u + B''u^7, & Sv^4 \frac{1}{M} = A''v^2 + B''v^{-6}. \end{array}$$

But for $u = 1$ the corresponding values of $v, \frac{1}{M}$ are

$$v = 1, -1, -1, -1, -1, -1,$$

$$\frac{1}{M} = 5, 1, 1, 1, 1, 1;$$

whence $A = A' = 10, A'' + B'' = 0$, or say the value of $S \frac{v^2}{M}$ is $A''u(1 - u^6)$.

The values of A'' are found very easily by the q -formula, viz. neglecting higher powers of q , we have

$$u = q^{\frac{1}{8}}\sqrt{2}, \quad v_0 = u^4\sqrt{2} \frac{1}{M_0} = 5, \quad v_1 = 2q^{\frac{1}{4}}\sqrt{2}, \quad \frac{1}{M_1} = 1;$$

hence

$$S \frac{v^2}{M} + S \frac{v^2}{M} = 5q^{\frac{1}{4}}(\sqrt{2})^4 + A''^{\frac{1}{4}}\sqrt{2},$$

that is, $A'' = 20$, and the equations are

$$S \frac{1}{M} = 10, \quad S \frac{v}{M} = 0, \quad S \frac{v^2}{M} = 10u^2, \quad S \frac{v^2}{M} = 10u^4, \quad S \frac{v^4}{M} = 20u(1 - u^6);$$

whence

$$\begin{aligned} F'v \cdot \frac{1}{M} &= 20u(1 - u^4) \\ &\quad - 10u^4(Su_4 - v) \\ &\quad - 10u(Sv_2u_1 + v^2Sv_0 - v^4) \\ &\quad - 10(Sv_3v_2u_1 + v^4Sv_2u_1 - v^4Sv_2 + v^2Sv_2 - v^4), \end{aligned}$$

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where Se_n , &c are the coefficients of the equation

$$v^5 + Se_nv^4 + Se_nv^3 - Se_nv^2 - Se_nv - u^4 = 0,$$

viz.

$$Se_n, \quad v_2v_1, \quad v_2v_2v_3, \quad v_3v_4v_5, \quad v_2v_3v_4v_5,$$

are

$$-4u^4, \quad +5u^6, \quad 0, \quad -5u^2, \quad 4u;$$

or the equation is

$$F'v \cdot \frac{1}{M} = 20u(1 - u^4)$$

$$\begin{aligned} & -10u^4(-4u^4 - v) \\ & -10u(4u + 5u^4v - 4u^2v^3 - v^4) \\ & -10(4u + 5u^4(v - v^2), \quad 0, \quad -5u^2, \quad 4u;) \end{aligned}$$

or, say

$$\frac{1}{2}F'v \cdot \frac{1}{M} = 5[v^4 + 4uv^4 + 6v^2u^4 + 4u^6v + vu^2(1 - u^2)],$$

where

$$\frac{1}{2}F'v = 3v^4 + 10vu^4 + 10v^4u^2 - 5vu^2 - 2u.$$

Hence also, reducing by the modular equation,

$$\frac{1}{2}vF'v \cdot \frac{1}{M} = 5v[v^4 + 4uv^5 + 6u^4v^2 + 2v(1 + u^2) + u^2],$$

the one of which forms is as convenient as the other.

71. Making the change $u, v, \frac{1}{M}$ into $v, -u, -5M$, we have

$$-\frac{1}{2}F'u \cdot 5M = 5[u^2 + 4u^4v + 6v^2u^4 + 4u^6v + 1 - v^2(1 - v^2)];$$

and comparing with the equation

$$\frac{v(1-v^2)}{u(1-u^2)} = -\frac{(1-v^2)uF'v}{(1-u^2)uF'u};$$

the modular equation is, in fact, found in the form

$$-5M(v - u^2)(1 - v^2) - (u - v^2)(1 - u^2) + (1 - u^2)(1 - v^2)5M = 0,$$

$$\text{i.e. } 5(uv - 1)5M = 4(u - v^2)(1 - u^2) + 5(uv - 1)(1 - v^2)5M = 0,$$

$$+5(1 - u^2)(1 - v^2)5M + 10uv(1 - uv)^25M = 0,$$

viz. from the value of $\frac{1}{M_n}$ for the case $n = 5$, given by Jacobi, the modular equation in its several forms, writing successively $u, v; u^2, v^2; u^4, v^4; u^4, v^4; u^4, v^4; = x, y$, are not curves of the same deficiency.

[p. 169]

Hence, replacing M, N by their values, this is

$$\begin{aligned} & -4uv(1 - v^2)(1 - u^2) \\ & -(u^2 - v^2)(1 - u^2 - u^2v^2 - u^2v^2 + u^2v^2 + u^2v^2 + v^4)(u^4 + 5u^2v + v^2) \\ & +4uv^4(u^2 + 5u^2 + u^2v^4)(1 - u^2v^4(u^2 - v^2) + u^2v^2) = 0; \end{aligned}$$

viz. writing $u^2 - v^2 = A, uv = B$, this is

$$\begin{aligned} & -4B(1 - u^2)(1 - v^2) \\ & -A(1 - A^2 - 5A^2B^2 - 3B^4)(A^2 + 5B + B^2) \\ & +4B(A^2 + A^2 + 5B)(1 - A^2B^2 - B^4) = 0, \end{aligned}$$

that is,

$$\begin{aligned} & -4B[(1 - A^2 - 5A^2B^2 - 2B^2 + B^4) \\ & -A(1 - A^2 - 5A^2B - 3B^2)(A^2 + 5B^2)(A^2 + 5B^2) \\ & +4B(A^2 + A^2 + 5B)(1 - A^2B^2 - B^4) = 0, \end{aligned}$$

viz.

$$\begin{aligned} & -4B(1 - A^2 - 5A^2B - 3B^2)(A^2 + 5B^2)(A^2 + 5B^2) \\ & -A(1 - A^2 - 5A^2B - 5B^2)(A^2 + 5B^2) = 0; \end{aligned}$$

or throwing out the factor $-(1 - A^2 - 5A^2B - 5B^2)(A^2 + 5B^2)$, this is

$$A(A^2 + 5B^2) + 4B(1 - B^2) = 0,$$

the modular equation, which is right.

The four forms of the modular equation, and the curves represented thereby.

Art. Nos. 72 to 75.

72. The modular equation for any value of n has the property that it may be represented as an equation of the same order $(n+1, n+1)$, the order being a prime, between u, v ; or between u^2, v^2 ; or between u^4, v^4 . As to this remark, that in general an equation $(\alpha, v)^{m,n} = 0$ of the order $2m$ between u, v may give an equation $(u^2, v^2)^{m,n} = 0$ of the order m between u^2, v^2 , the required equation is in u^2 and v^2 . But if we have $\phi(u, v) = 0$ of the order $2m$ between u, v ; and the like passing to the inverse equation $\phi(u, v) = 0$, if the coefficients reduce itself to an equation $(u, v)^{m,n} = 0$ in u, v , the curve in (u, v) is of the order $2m$, that is, of the same deficiency as the original curve of the order $2m$ in (u, v) . The transformed curve in $x, y, = u^2, v^2$ would in like manner be of the order $2m$, of the same deficiency as the original curve; and in particular the curves represented by the equation in their final its four several forms, writing them successively $u, v; u^2, v^2; u^4, v^4; u^4, v^4; = x, y$, are not curves of the same deficiency.

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would correspond 8 values of u , therefore 8a values of v , giving the same number, 8. values of v^2 ; that is, the values of u^2 correspond to a given value of u^2 . There is, in fact, no group themselves in eights cusps at infinity, the coefficients are only two coefficients α, β , which are also the last but one and last coefficients ρ, σ . Hence, from the values of $\alpha, \beta, \rho, \sigma$, respectively, are derived herefrom as follows; viz. and u^4, v^4 respectively, are derived herefrom as follows; viz. and

I. $y^4 - x^4 + 2xy(x^2y^2 - 1) = 0;$

and it is at once seen that the remaining forms wherein x, y denote $u^2, v^2; u^2, v^2; u^4, v^4$ respectively, are derived herefrom as follows:

II. $(y^2 - x^2)^4 - 4xy(xy - 1)^4 = 0,$
or $x^4 + 6x^2y^2 + y^4 - 4xy(x^2y^2 + 1) = 0;$

III. $(y^2 + 6xy + y^2)^2 - 16xy(xy + 1)^2 = 0,$
or $x^4 + 6xy + y^4 - 16xy(4x^2y^2 - 3xy) \cdot 32x^4y^4 - 96x^3y^3 - 70x^2y^2 - 96xy + 64 = 0;$

IV. $(y^4 + 6xy + y^4)^4 - 16xy(4x^2y^2 - 3xy) \cdot 32x^4y^4 - 96x^3y^3 - 70x^2y^2 - 96xy + 64 = 0;$

where the subscript line, showing in each case what the equation becomes on writing therein $x = 1$, serves as a verification of the numerical values.

The curve IV. being unicursal, the coordinates may be expressed rationally in terms of a parameter; in fact, we have

$$x = \frac{a^4(2+a)}{1+2a}, \quad y = \frac{a(2+a)^4}{(1+2a)^4}.$$

These values give

$$\begin{aligned} 16xy &= 16a^4(2+a)^4, \quad +(1+2a)^4, \\ 4+4xy-3x-3y &= (4, 8, 12, 32, 50, 32, 12, 8, 4)(1, a)^4, \quad +(1+2a)^4, \\ x^4+6xy+y^4 &= 4a^4(2+a)^4(4, 8, 12, 32, 50, 32, 12, 8, 4)(1, a)^4, \quad +(1+2a)^4, \end{aligned}$$

and the equation of the curve is thus verified.

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75. The foregoing equations may be exhibited in the square diagrams:—

[Tables: four 5×5 square diagrams labelled I, II, III, IV, expressing the non-zero entries of the modular polynomials in the variables $y^4, y^3, y^2, y, 1$ along the top axis and $x^4, x^3, x^2, x, 1$ down the left axis. Each diagram is read by multiplying out, with the row-sums giving the coefficients of the modular polynomial in y . The accompanying labels at the foot give the corresponding modular form $(y-1)^k(y \pm 1)^k$, and so on.]

Diagram I.

	y^4	y^3	y^2	y	1
x^4					-1
x^3		$+2$			
x^2					
x			-2		
1	$+1$				

$$1 \quad +2 \quad 0 \quad -2 \quad -1 = (y+1)^2(y-1);$$

Diagram II.

	y^4	y^3	y^2	y	1
x^4					$+1$
x^3			-4		
x^2			$+6$		
x		-4			
1	$+1$				

$$1 \quad -4 \quad +6 \quad -4 \quad +1 = (y-1)^4.$$

where the subscript line, showing in each case what the equation becomes on writing therein $x = 1$, serves as a verification of the numerical values.

The curve IV., being unicursal, the coordinates may be expressed rationally in terms of a parameter, in fact, we have

$$x = \frac{a^4(2+a)}{1+2a}, \quad y = \frac{a(2+a)^4}{(1+2a)^4}.$$

These values give

$$16xy = 16a^4(2+a)^5, \quad +(1+2a)^5,$$

$$4 + 4xy - 3x - 3y = (4, 8, 12, 32, 50, 32, 12, 8, 4)(1, a)^8 \quad + (1+2a)^8,$$

$$x^4 + 6xy + y^4 = 4a^4(2+a)^5(4, 8, 12, 32, 50, 32, 12, 8, 4)(1, a)^8 \quad + (1+2a)^8,$$

and the equation of the curve is thus verified.

[p. 172] 76.

Considering in like manner the modular equation for the quintic transformation, we derive the four forms as follows:—

I. $x^2y^2 + 5xy(x-y) + 4xy(1-x^2y^2) = 0;$

II. $[x^2 - y^2 - 5xy(1 - x^2y^2)]^2 - 16xy(8 - 5x^2 - 5y^2 + 10x^2y^2 - 5xy)xy = 0$, that is, $x^4 + 15xy + 15xy + y^4 - 4xy(8 - 5x^2 + 10xy^2 - 5y^2 + 5xy^2) = 0;$

III. $(x^4 + 655xy + 655x^4y + y^4 - 4xy(8 - 5x^4 + 10x^4y^4 - 5y^4 + 5xy^4) - 640xy = 640x^4y^4);$
 $+xy(-256 + 320v^2 + 320y^2 - 70x^2y^2 - 70y^2 - 256x^2y^2) - 640xy(x^4y^4 - 256x^2y^2 - 256) = 0;$

IV. $(x^4 + 655xy + 655x^4y + y^4 - 640xy - 640x^4y^4)$
 $-xy(-256 + 320x^2 + 320y^2 - 70x^2y^2 - 70x^2y^2 - 256y^2) + 320xy(x^4y^4 - 256x^2y^2 - 256) = 0;$

or, expanding the two terms in the last equation separately, this is

[Table: a square diagram extending to powers $x^i y^j$ with i, j up to 6, listing the non-zero entries of the polynomial in question. Cayley presents the diagram in a single 7×7 block: the column heads are $x^6, x^5, x^4, x^3, x^2, x, 1$ and similarly for the rows. The numerical entries reproduced from the printed diagram are:]

	x^6	x^5	x^4	x^3	x^2	x	1	= 0.
xy							−63536	
y^2						+163840		
y^3				+409600			−543720	
y^4			−1280		−138240			
y^5		−538400			−138240		+138240	
y^6	+1280		−538400	−639400		+44800		

[Cayley then expands the second of the equations of § 75 (the modular equation for the quintic transformation in its second form). The full table covers the next two pages and is presented as a single 7×7 array; the reproduction here is as in the original; entries omitted in the original print remain blank in our table.]

[p. 173] 77. The square diagrams are:—

Diagram I.

	y^5	y^4	y^3	y^2	y	1
x^5						−1
x^4		+4				
x^3			−5			
x^2		+5				
x			−4			
1	+1					

1 + 4 + 5 0 − 5 − 1 = $(y + 1)^4(y - 1)$.

Diagram II.

	y^5	y^4	y^3	y^2	y	1
x^5						+1
x^4		−16			+10	
x^3			+15			
x^2			−20			
x		+10			−16	
1	+1					

1 − 6 + 15 − 20 + 15 − 6 + 1 = $(y - 1)^6$.

Diagram III.

	y^5	y^4	y^3	y^2	y	1
x^5						+1
x^4	−256		+320		−70	
x^3		−640		+655		
x^2	+320		−600		+320	
x		+655		−640		
1	−70		+320		−256	

1 − 6 + 15 − 20 + 15 − 6 + 1 = $(y - 1)^6$.

Diagram IV.

	y^5	y^4	y^3	y^2	y	1
x^5						+1
x^4	−65536	+163840	−138240	+43520	−3590	
x^3	+163840	−133120	−207360	+153135	+43520	
x^2	−138240	−207360	+691180	−207360	−138240	
x	+43520	+153135	−207360	−133120	+163840	
1	−3590	+43520	−138240	+163840	−65536	

$$1 \quad -6 \quad +15 \quad -20 \quad +15 \quad -6 \quad +1 = (y-1)^6.$$

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where the subscript line, showing in each case what the equation becomes on writing therein $x = 1$, serves as a verification of the numerical values.

78. The curve I. has at the origin a dp in the nature of a flecnod, viz. the two branches are given by $x^2 + 4y = 0$, $x - 4y^4 = 0$ respectively; and there are two singular points at infinity on the two axes respectively, viz. the infinite branches are given by $-y - 4u^2 = 0$, $x - 4y^4 = 0$ respectively. Writing the first of these in the form $-yx^2 - 4u^2 = 0$, $x - 4y^4 = 0$ respectively. Writing the first of these in the form $-yx^2 - 4u^2 = 0$, the point at infinity on the axis $x = 0$ is ∞ dps*.

Moreover, as remarked to me by Professor H. J. S. Smith, the curve has 8 other dps; viz. writing ω to denote an eighth root of -1 , ($\omega^4 + 1 = 0$), the eight branches are two singular points at infinity on the two axes respectively, viz. the infinite branches are given by $-y - 4\omega^4 = 0$, $x - 4\omega^4 = 0$ respectively. Writing the first of these in the form

$$\begin{array}{rcl} 6x^4 & = & +6, \\ +20xy^2 & - & 20, \\ -10xy^4 & - & 10, \\ +4y & + & 4, \\ -20x^4y^4 & + & 20; \end{array} \quad \begin{array}{rcl} -6y^2 & = & +6 \\ +10xy & - & 10 \\ -20xy^2 & - & 20 \\ +4x & + & 4 \\ -20x^4y^4 & + & 20; \end{array}$$

or these functions become $= 0$, in like manner $x = \omega$, $y = -\omega$, and that the curve is a $\frac{4}{2}$ dps; viz. instead of these, we have $1+12+2+6+6 = 21$ dps.

In all $1+12+2+6+6+12 = 33$ dps.

And in IV. we thence have $1+12+3+6 = 21$ dps; and, besides, the 12 dps given by

$$x^2 + 5xy^2 + y^2 = 0, \quad 8 - 5x^2 + 10xy - 5y^2 + 5xy^4 = 0,$$

($= 0$, $y = 0$); we have besides 12 more, which form the 12 cusps of the figure $5x = 0$, $= 296 + 320x - 70x^4 - 660xy + 70x^4y^2 - 320xy^4 - 256x^4y^4 = 0$, ($296 + 820x - 70x^4 - 660xy + 70x^4y^2 - 320xy^4 - 256x^4y^4 = 0$), (these intersect in $1+12+1+6+6+6 = 33$ points); there are 12 dps, $10 - 4 = 12$ intersections); there are $1+12+1+6+6+6 = 33$ dps.¹

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Arranging the results in a tabular form and adding the values of the deficiency, we have

	dps.	Def.
I.	$1 + 12 + 8 = 21$,	$= 15$,
II.	$1 + 12 + 4 + 12 = 29$,	$= 7$,
III.	$1 + 12 + 2 + 6 + 12 = 33$,	$= 3$,
IV.	$1 + 12 + 1 + 5 + 6 + 12 = 35$,	$= 1$,

¹* These curves follow from the general theorem in the paper "On the Higher Singularities of Plane Curves," *Camb. and Dubl. Math. Journ.*, t. vi. (1865), pp. 127–225, [271]; but they may be at once seen to be true from the consideration that the curve $yx^2 - 2u = 0$, which has only the singularity in question, is unicursal; the singularity is then $\infty - 1$ dps.

so that the curve IV. is a curve of deficiency 1, or bicursal curve. It appears by Jacobi's investigation for the quintic transformation (*Fund. Nov.* pp. 26–28, [*Ges. Werke*, t. i., pp. 77–79]) that we can in fact express x, y , that is, u^4, v^4 , rationally in terms of the parameters α, β connected by the equation

$$\alpha^2 = 2\beta(1 + \alpha + \beta),$$

which is that of a general cubic (deficiency=1); in fact, we have

$$\frac{2-\alpha}{\alpha-2\beta} = \frac{v^2}{u^2}, \quad \beta = \frac{u^2}{v},$$

that is,

$$u^4(=x) = \beta^4 \left(\frac{2-\alpha}{\alpha-2\beta} \right), \quad v^4(=y) = \beta^4 \left(\frac{2-\alpha}{\alpha-2\beta} \right)^4,$$

where α, β satisfy the relation just referred to. The actual verification of the equation IV. by means of these values would not be without some labour.

79. In the general case of any odd prime, the equation in u as the origin dp (in the nature of a fleshnode) and an infinity singular point with singular points in dps $\frac{1}{2}(n-1)(p-3)$ dps; and there is a result obtained by Professor Smith, that there are besides $\frac{1}{2}(n-1)(p-3)$ dps; but I have not investigated the nature of these. And the Table of dps and deficiency then is

	dps.	Def.
I. $1 + (p-1)(p-2) + (p-3)$	=	$2p^2 - 7p + 6, 4p - 5,$
II. $1 + (p-1)(p-2) + \frac{1}{2}(p-3) + \frac{1}{2}(p-1)$		$2p^2 - 5p + 0, 3p - 3,$
III. $1 + (p-1)(p-2) + \frac{1}{2}(p-3) + \frac{1}{2}(p-1) + \frac{1}{2}(p-1) + \frac{1}{2}(p-1)$		$2p^2 - 4p + 0, p - 2,$
IV. $1 + (p-1)(p-2) + \frac{1}{2}(p-3) + \frac{1}{2}(p-1) + \frac{1}{2}(p-1) + \frac{1}{2}(p-1)$		$2p^2 - 4p + \frac{1}{2}, \frac{1}{2}p - \frac{1}{2};$

viz. the values of the deficiencies being as in the last column, the total number of dps must be as in the last but one column.

579.

Address Delivered by [Professor Cayley as] the President of the Royal Astronomical Society on Presenting the Gold Medal of the Society to Professor Simon Newcomb.

[From the *Monthly Notices of the Royal Astronomical Society*, vol. XXXIV. (1873—1874), pp. 224—233.]

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The Council have awarded the medal to Professor Simon Newcomb for his Researches on the Orbits of *Neptune* and *Uranus*, and for his other contributions to mathematical astronomy. And upon me, as President, the duty has devolved of explaining to you the grounds of their decision.

I think it right to remark that it appears to me that, in the award of their highest honour, the Council of a Society are not bound to institute a comparison between heterogeneous branches of a science, or classes of research—to weigh, for instance, mathematical against observational astronomy or astronomical physics; or, in the several branches respectively, the happy idea which originates a theory against the patience and the skilled labour which develope and carry it out; and still less to decide between the merits of different workers in the science. It is enough that the different branches of a science coming before them in different years, the medal should in every case be bestowed as a recognition of high merit in some important branch of the science.

Before speaking of the Tables, I will notice some of Professor Newcomb's other works.

MEMOIR "On the secular Variations and mutual Relations of the Orbits of the Asteroids," *Mem. American Academy*, vol. v. (1860), pp. 124—152. The object is to examine those circumstances of the forms, positions, variations, and general relations of the asteroid orbits which may serve as a test, complete or imperfect, of any hypothesis respecting the cause from which they originated, or the reason why they are in a

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group by themselves. Every *a posteriori* test is founded on the supposition, that the hypothesis necessarily or probably implies that certain conditions must be satisfied by the asteroids or their orbits, viz. in the one case the conditions are those which follow necessarily and immediately from the hypothesis itself, in the other case those which are deducible from it by the principle of random distribution. The two principal hypotheses are that of Olbers, where the asteroids are supposed to be the fragments of a shattered single planet; and the hypothesis that they were formed by the breaking up of a ring of nebulous matter. On the first hypothesis the orbits of all the asteroids once intersected in a common point; the second affords no conclusion equally susceptible of an *a posteriori* test.

But for a rigorous or probable test of either hypothesis, what is needed is rigorous expressions in terms of the time for the eccentricity, inclination, and longitudes of perihelion and node of each of the asteroids considered, or, what is the same thing, the computation of the secular variations of the quantities h , l , p , q , which replace these elements. The investigation is applied to these asteroids the elements of which were determined with sufficient accuracy, and the eccentricities and inclinations of which were sufficiently small (limit taken is 11°). And the backbone of the memoir is a shattered single planet: and the hypothesis that they were formed by the breaking up of a ring of nebulous matter. It is seen that the orbits of all the asteroids once intersected in a common point: the second affords no conclusion equally susceptible of an *a posteriori* test.

“Investigation of the Distance of the Sun and of the Elements which depend upon it, from the Observations of *Mars* made during the Opposition of 1862, and from other Sources,” *Washington Observations for 1865*, Appendix II., pp. 1—29. The chief part of this valuable Memoir is occupied with a determination of the solar parallax by the discussion of the observations of *Mars* made in 1862 on the plan of Winnecke; three partial discussions had previously appeared, but those having been by comparisons of pairs of observations, one in each hemisphere, many observations in one hemisphere were lost by want of a corresponding observation in the other hemisphere; and out of a total of nearly 300 observations, only 125 were utilised. The idea is, the perturbations of the Earth and *Mars* being perfectly known for the period under consideration, every observation of the planet would lead rigorously to an equation of condition between its parallax, the six elements of its orbit, and the six elements of condition between its parallax, the six elements of its orbit, and the six elements of

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the Earth’s orbit—thus 13 or more observations, when compared with any theory, should suffice to correct the errors of that theory. But the observations extending only over a short interval, say one month, the coefficients would be so minute as to give no trustworthy value of the corrections; the equations only suffice to determine a few functions of the elements which, being determined, the equations will be satisfied by widely differing values of the elements, if only those values are such as to give to the functions their right values. And by fixing *a priori* the entire number of functions in question, and using them in place of the elements of the Earth and *Mars*, the equations will be practically as rigorous as if all the 13 unknown quantities had been introduced. By such considerations as those, each observation is made to give a relation between only 3 unknown quantities, the correction of the Sun’s parallax being one of them.

The principle appears to be one of extended application, in regard to the proper mode of dealing with the constantly recurring problem of the determination of a set of corrections from a large number of linear equations; and it is used by the author in regard to the equations which present themselves in his theories of *Neptune* and *Uranus*.

Returning to the *Mars* observations, these were made at six Northern and three Southern Observatories, the total number being 154 Northern, and 143 Southern, together 297 observations. There was the difficulty of reducing to a concordant system the observations at the different Observatories, since (the whole number of comparison stars not being observed on each night) the adopted mean position of each of them was not unimportant. But this being carefully discussed and allowed for, the observations, extending from August 21 to November 3, 1862, are divided into five groups, and from these is deduced a correction to the provisional value $8''.9$ of the parallax. The author then reproduces or discusses other determinations, from micrometric observations of *Mars*, the parallactic inequality of the Moon, the lunar equation of the Earth, the transit of 1769, and Foucault’s experiment on light—the last result, as not a strictly astronomical one,

and with no means of assigning its probable error, is left out of consideration—and the combination of the remaining ones gives the author's concluded value of the parallax; from which other astronomical constants are deduced.

“On the Right Ascensions of the Equatorial Fundamental Stars and the Corrections necessary to reduce the Right Ascensions of different Catalogues to a mean homogeneous System,” *Washington Observations for 1870*, Appendix III., pp. 1—73.

This important Memoir is referred to in the Council Report for 1873. The object is to do for the right ascensions of the equatoreal and zodiacal Stars what had been done by Auwers for the declinations, namely, to furnish the data necessary to reduce the principal original catalogues of stars to a homogeneous system by freeing them of their systematic differences. The results are contained in two tables of corrections (as depending on the R.A. and N.P.D. respectively) to the several catalogues; and in a table of concluded mean right ascensions for the beginning of each fifth Besselian year,

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1750 to 1900, of 32 fundamental Stars, and of periodic terms in the right ascensions of *Sirius* and *Procyon*.

The evil of systematic differences between the observations of different Observatories of course presents itself in every case where such observations have to be combined: for instance, in the just-mentioned determination of the solar parallax by the observations of *Mars*; and in the making of a set of planetary tables; and all that tends to remove or diminish it is most important to the progress of Astronomy. I cannot help thinking that there should be some confederation of Observatories, or Central calculating Board, for publishing the lunar and planetary observations, &c., reduced to a concordant system. It seems hard upon the maker of a set of planetary tables that he should not at least have, ready to hand for comparison with his theory, a single and entire series of observations of the planet for the comparison period.

“Théorie des Perturbations de la Lune, qui sont dues à l'action des Planètes,” *Liouville*, t. XVI. (1871), pp. 1—45. This is a very important theoretical Memoir on the disturbed motion of three bodies: a problem which, notwithstanding the limited extent of the perturbations considered at all, has nevertheless to be considered with a different mode of dealing with the constantly recurring problem of the determination of a set of corrections from a large number of linear equations; and it is used by the author in regard to the equations which present themselves in his theories of *Neptune* and *Uranus*.

“Théorie des Perturbations de la Lune, qui sont dues à l'action des Planètes,” *Liouville*, t. XVI. (1871), pp. 1—45. The evil of systematic differences between the observations of different Observatories of the present Memoir, Professor Newcomb starts from this problem in a fit more actually solved, viz. he takes the coordinates of the three bodies, Sun, Earth, and Moon) as given in terms of 18 constants of integration*. And then considering the equations of the system as the action of a planet, represented by the means of a disturbing function, he applies to the system of the three bodies the method of the variation of the elements. The six elements which determine the motion of the centre of gravity of the system are left out, and not considered; there remain to be considered 12 elements only (instead of being as before, with the elements *a posteriori*, retained), six of these belong to the relative orbit of the Earth, l_1 , l_2 , &c., are functions the invariation of which is a leading step in the theory, and they are also functions of the inclination of which is a leading step in the theory, and they are also the expressions of the last-mentioned six functions are, it is stated, be formed with facility by means of the developments (obtained therein from the lunar theory) of the rectangular coördinates x , y , z , as periodic functions of the time. With these twelve elements, the expressions for the variations assume the canonical form

$$\frac{dx_1}{dt} = \frac{dR}{de_1}, \quad \frac{de_1}{dt} = -\frac{dR}{dx_1}, \quad \&c.$$

The concluding part of the Memoir contains approximate calculations which seems to show that the whole process is in very practicable one: but the author remarks that it is only proper to defray justice to Delaunay to say that, starting from his (Delaunay's) final differential equations, and regarding the planet as adding new terms to the disturbing function, then by introducing the laws of the same degree of rigour as to the disturbing opinion as to the comparative facility of methods; but irrespectively of the possible applications of the method, the Memoir is, from the boldness of the conception and beauty of the results, a very remarkable one, and constitutes an important branch of Theoretical Dynamics*.

I come now to the planets *Neptune* and *Uranus*: it is well-known how, historically, the two are connected. The increasing and systematic inaccuracies of Bouvard's Tables of *Uranus* were found to be such as could not be accounted for by the existence of differential equations, and regarding the planet as adding new terms to the disturbing function, he could have obtained results of the same degree of rigour as to the disturbing forces, leading to Hansen's first inequality produced by *Venus*.

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coördinates x, y, z , as periodic functions of the time. With these twelve elements, the expressions for the variations assume the canonical form

$$\frac{dx_1}{dt} = \frac{dR}{de_1}, \quad \frac{de_1}{dt} = -\frac{dR}{dx_1}, \quad \&c.$$

The concluding part of the Memoir contains approximate calculations which seems to show that the whole process is in very practicable one: but the author remarks that it is only proper to defray justice to Delaunay to say that, starting from his (Delaunay's) final differential equations, and regarding the planet as adding new terms to the disturbing function, he could have obtained results of the same degree of rigour as to the disturbing forces, leading to Hansen's first inequality produced by *Venus*.

Everything in the Lunar Theory is laborious, and it is impossible to form an opinion as to the comparative facility of methods, the irrespectively of the possible applications of the method, the Memoir is, from the boldness of the conception and beauty of the results, a very remarkable one, and constitutes an important branch of Theoretical Dynamics*.

I come now to the planets *Neptune* and *Uranus*: it is well-known how, historically, the two are connected. The increasing and systematic inaccuracies of Bouvard's Tables of *Uranus* were found to be such as could not be accounted for by the existence of differential equations, and regarding the planet as adding new terms to the disturbing function, then by introducing the laws of the same degree of rigour as to the disturbing opinion as to the comparative facility of methods; but irrespectively of the possible applications of the method, the Memoir is, from the boldness of the conception and beauty of the results, a very remarkable one, and constitutes an important branch of Theoretical Dynamics*.

[The Address continues beyond p. 180; the remaining pages 181–200 of the present range belong to it.]